Social networks and experienced inequality

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Abstract. Traditional measures of inequality, such as the Gini coefficient, involve pairwise comparisons across all members of a given population. But, most people possess information about, and therefore experience inequality in comparison to, only a subset of the population. In this paper, we provide simple axioms to describe inequality as experienced in social networks. We propose an index to measure aggregate experienced inequality that is consistent with these axioms. We then compute the Gini coefficient and 'experienced inequality' in 75 villages in Karnataka, India. We show that for a given wealth distribution, the social network could either accentuate or diminish experienced inequality. We show, first analytically and then empirically, how this can happen with respect to two network properties. Firstly, wealth-based homophily is negatively associated with experienced inequality. Secondly, caste-based homophily is negatively associated with experienced inequality when within-caste inequality is less than the overall Gini coefficient (and positively associated when the opposite is true).

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> It is clear also what kind of people we envy... we envy those who are near us in time, place, age, or reputation. ...we do not compete with men who lived a hundred centuries ago, or those not yet born, or the dead, or those who dwell near the Pillars of Hercules, or those whom, in our opinion or that of others, we take to be far below us or far above us. So too we compete with those who follow the same ends as ourselves: we compete with our rivals in sport or in love, and generally with those who are after the same things. - Aristotle (Rhetoric, Book 2 Chapter 10)

1. Introduction

In this paper we explore the measurement of *experienced inequality*. Inequality can be understood in two ways. The first is as the degree of spread or variance between all outcomes in an underlying distribution. In this interpretation, it is purely a statistical property of the distribution. Second, most standard inequality measures have a further interpretation as an aggregation of the differences between outcomes for all individuals (Berrebi and Silber 1985). The traditionally used Gini coefficient, for example, is measured as half the sum of pairwise differences across all individuals in a society, normalized by mean income and population. As such, it is a transformation and summation of each individual's differences with all members of society.

Adopting this individual based interpretation of inequality measurement allows us to assess the salience of inequality for human behaviour and for societal outcomes. Thus for example, following from Runciman (1961), inequality statistics have often been motivated as an aggregate measure of relative deprivation, and as experienced distance from income derived by all individuals. Later measures of distribution such as polarization (Esteban and Ray 1994) also take this individual interpretation, suggesting that it is the outcome of individual identification and alienation from all others.

Commonly deployed inequality indices such as the Gini coefficient or the Atkinson and Theil indices take into account every individual's relationship with every other individual to obtain an aggregate of every such comparison. But, in real world situations, people do not typically come in contact with or possess information about every member of the population. Individuals interact with, and get information from, a subset of the population. It is these interactions and this information that provides motivation for their behavior. Aristotle's description above of envy as a socially embedded phenomenon – as something relational, defined with respect to a well defined subset of the population – illuminates this intuition well. Research across a variety of disciplines - economics, social psychology, anthropology and sociology, for example - confirms the salience of local, as opposed to universal, comparisons.

In economics, the concept of Veblen effects - the idea that human beings are motivated by the actions of *relevant* others - is well known and empirically verified in a variety of instances. Thus for example societal work habits or consumption can depend on comparisons between individuals and others (Bowles and Park 2005; Frank 2005; Veblen and Mills 1899). Similarly, evidence suggests that higher-income individuals are less generous if they already reside in a highly unequal area (Côté, House, and Willer 2015; Nishi and Christakis 2015). Social comparison theory (Festinger 1954) has been used to explore a wide variety of contexts in which disparities with particular others lead to changed behaviour. For example, in educational psychology, grade performance of students is informed by the performance of close friends and peers (Jansen, Bosa, and Lorenz 2022). Evidence from political science shows that political opinions and attitudes towards redistribution are mediated through actually experienced disparities. Minkoff and Lyons (2019) suggest that experienced neighborhood inequality affects people's perception of the gap between the rich and the poor, and these may have strong effects on preferences towards redistribution. Local relative disparity has also been shown to play a significant role in international migration (Stark and Taylor 1989). Finally, experienced inequality in terms of status and rank order in small groups are associated with significant health outcomes in both humans and a variety of animal species (Sapolsky 2004).

Experienced inequality ($\mathbb{E}$), as we define it, is therefore, disparity as mediated through the social configurations that an individual faces. This differs from standard inequality measures which presume that comparisons are universal across the society. As a simple example of how $\mathbb{E}$ may diverge from standard inequality measures, imagine two sub-populations of a society that are internally very equal, and which never interact with each other or do not possess information about each other, but which are at very different levels of income. Standard measures would likely evaluate inequality to be very high in this instance, but individuals do not possess information nor interact across groups and therefore may not be *experiencing* the overall high inequality.

A natural way to model these differences is to look at both the distribution of income as well as the network of connections between individuals. Bowles and Carlin (2020) reformulate the Gini coefficient as a measure of experienced inequality on a complete social network. Their reformulation ensures that unlike the conventionally used Gini coefficient, this measure does not have a downward bias for smaller societies[1]. It is also a more desirable alternative to the traditional Gini since it lends itself to real-world situations where social and economic networks are not complete.

More recently, Stark, Bielawski, and Falniowski (2023) propose an index of relative deprivation on social networks. While inequality can be understood as deprivation or envy, in this paper, we do not privilege any particular interpretation of difference, but instead treat it as an

[1]For further discussion on the properties and implications of different formulae for the Gini coefficient, see Bowles and Carlin (2023) as a part of a larger discussion with Debraj Ray and Rajiv Sethi in Ray et al. (2021).

information set for individuals, appropriately aggregated. We can imagine a situation for example, where two unequal individuals who were previously unconnected are now linked. While it is true that the poorer person is experiencing a deprivation, it is not clear that this need necessarily be a social bad, since it is possible that the person now has new information to evaluate and improve her situation. Similarly, the wealthier person, too, experiences an expansion in her information set.

We also contribute to a set of studies that attempt to generalise standard metrics to the context of networks. For example Coscia (2020) proposes a generalised Euclidean distance for vectors on networks. Hohmann, Devriendt, and Coscia (2023) present a metric of polarisation on networks, and Devriendt, Martin-Gutierrez, and Lambiotte (2022) develop a way to measure variance and covariance on networks taking into account the network distances between nodes.

Taking off from this point, we explore experienced inequality on empirically observed networks. In what follows, we begin by providing an axiomatic basis for thinking about experienced inequality and suggest an index that satisfies these axioms. Experienced inequality, as we conceive and measure it, can be affected by both the wealth distribution and the network structure. Consider Figure 1, drawn from a real-life village based network that we more fully describe shortly. The figure depicts two different networks. Panel (A) plots the relative (or kinship) network where an individual is connected to more people like themselves. Panel (B) plots another network (the help-decision network) which links people with those they would reach out to for help with decision making. Each individual's wealth is the same in both networks, but in the second network, individuals may be naming those of higher status. And so, we find that even though the Gini coefficient is the same, experienced inequality ($\mathbb{E}$) is greater in the second network than in the first.

To measure the role of the network structure in either exacerbating or dampening the experience of a given wealth distribution, we use the ratio of experienced inequality to the Gini coefficient. We call this ratio the network multiplier of the Gini, r. We then explore experienced inequality with respect to two network properties. We underline two facts for any given wealth distribution. First, experienced inequality is negatively associated with wealth-based homophily in a network. Second, (b) group-based homophily is negatively associated with experienced inequality when average within-group inequality (measured as the sum of the within group Gini coefficients weighted by the wealth shares) is less than the overall Gini coefficient.

We then compute experienced inequality in 75 villages in Karnataka, India. We find that the the Gini coefficient overestimates the inequality experienced in 25 villages and underestimates it in 50 villages. We then empirically verify the analytical results with respect to homophily mentioned above. We find experienced inequality is indeed negatively associated with wealth-based homophily in these networks. We also find that caste-based homophily is negatively associated with experienced inequality when the average within-caste inequality is less than the overall Gini coefficient.

2. Measuring experienced inequality

There have been a number of studies looking at societal parameters that derive from relative comparisons between individuals. Runciman (1961), Yitzhaki (1979), Stark, Taylor, and Yitzhaki (1986), and Stark, Bielawski, and Falniowski (2023) suggest that for an individual, an appropriate psychological interpretation of interpersonal comparison is that of relative deprivation or disparity. This supposes that any individual compares himself or herself to a reference group and feels a disparity if there is anybody with a higher level of the good. Cowell and Ebert (2008) derive two different indices – one relative and one absolute – to aggregate measures of envy. Specifically, the measure draws from the distance of each person's wealth from his or her immediately richer neighbour. An even earlier attempt to aggregate specific distances is due to Podder (1996) who uses relative disparities to derive an aggregate measure of felt disparity. An interesting aspect of his analysis is that relative deprivation is maximized when half the population is comparing itself to the other half.

Bowles and Carlin (2020) reformulate the Gini coefficient as a measure of experienced inequality on a complete social network.

$$G = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} |x_i - x_j|}{\frac{n(n-1)}{2} 2\bar{x}} \tag{2.1}$$

where $\bar{x}$ is the mean wealth and $\frac{n(n-1)}{2}$ is the total number of unique connections present on a complete social network. This can be simply interpreted as an aggregation of disparities between an individual's income or wealth and that of all others, appropriately normalized. This measure advances a network based interpretation of inequality as experienced by individuals.

In what follows, we explore experienced inequality on real world social networks that are often not complete. We define some axioms regarding experienced inequality that will provide the basis for an index.

2.1. Individual experienced inequality. Let $\mathbf{x}$ be a vector of wealth of n individuals. Let $\mathbf{e}$ be a vector of length n with each element e_i representing the inequality experienced by individual i.

Let $\mathbf{D}$ be an $n \times n$ matrix whose element $d_{ij} \geq 0$ gives the distance between x_i and x_j. For example, we could have $d_{ij} = |x_i - x_j|$.

A1: *The inequality experienced by an individual will be non-decreasing in the individual's distance in wealth from each other individual in society.*

That is,

$$\frac{\partial e_i}{\partial d_{ij}} \geq 0 \tag{2.2}$$

Now, let the $n \times n$ matrix $\mathbf{A}$ have elements $a_{ij} \geq 0$ that denotes the strength of the social connection of individual j to i.

A2: *The effect on the inequality experienced by individual i of its wealth distance from individual j is higher if the social connection of individual j to i is stronger.*

That is,

$$\frac{\partial}{\partial a_{ij}} \left(\frac{\partial e_i}{\partial d_{ij}} \right) \geq 0 \tag{2.3}$$

A simple additive measure that satisfies these two axioms would be

$$e_i = \sum_{j \neq i} a_{ij} d_{ij} \tag{2.4}$$

A3: **Neighbourhood population invariance**: *The inequality experienced by an individual should be normalised by the number of people with which they are doing comparisons, i.e. for individual i, this should be normalised over the number of individuals $j \neq i$ where $a_{ij} > 0$.*

The experienced inequality for an individual should be invariant to replications of any neighbors. We impose this to avoid a version of what Parfit (1984) refers to as the 'repugnant conclusion'. Specifically, imagine two neighbours with an arbitrary small difference between them. Now imagine that the neighbor is replicated a large number of times, so that there is now a large population of individuals all arbitrarily close to each other. Now consider the same two neighbors with a very large difference between them. Imposing the axiom ensures that the large population with virtually no difference between members is not seen to have more inequality than the minimal two person society with a vast difference.

Therefore, the measure of individual experienced inequality becomes

$$e_i = \frac{\sum\limits_{j \neq i} d_{ij} a_{ij}}{\sum\limits_{j \neq i} \mathbb{I}_{a_{ij} > 0}} \tag{2.5}$$

where $\mathbb{I}_{a_{ij}>0}$ is an indicator function that takes the value 1 when $a_{ij} > 0$.

To fix ideas, let us assume a simple measure of wealth-distance as the absolute difference i.e. $d_{ij} = |x_i - x_j|$. Similarly, if we assume all individuals to be connected by an unweighted, undirected social network, then we can set $a_{ij} = \gamma^{b_{ij}-1}$ where $\gamma \in (0,1]$ signifies the fidelity of information transmission through the network, and b_{ij} is the shortest network path between nodes i and j. Therefore, if i and j are immediate neighbours i.e. $b_{ij} = 1$, then $a_{ij} = 1$. And if $\gamma < 1$, then as b_{ij} increases, a_{ij} tends to 0 .

This gives us the following measure of the experienced inequality of individual i as

$$e_i = \frac{\sum_{j \neq i} |x_i - x_j| \gamma^{b_{ij}-1}}{\sum_{j \neq i} \mathbb{I}_{\gamma^{b_{ij}-1} > 0}} \tag{2.6}$$

In traditional measures of inequality that start from the notion of interpersonal comparison, it is as if every individual has connections to and full information about the income of every other individual. Both these assumptions – full connectedness and full information – are highly restrictive. In reality, individuals have only a bounded set of comparators and incomplete information about the income/wealth of these. Furthermore these comparators may be those who are directly connected, or connected through other nodes. In our measure, social networks not being complete allows us a more accurate representation of the comparator group. Additionally, the parameter γ allows us to relax the assumption of full information.

In a fully connected network, if $\gamma = 1$ (i.e. there is no decay of information at all through the network) and every individual knows every other individual perfectly well, the individual experienced inequality is

$$e_i = \frac{\sum_{j \neq i} |x_i - x_j|}{n-1} \tag{2.7}$$

This pairwise comparison is the basis for the Gini as found in Bowles and Carlin (2020).

If there is no information transmission beyond the immediate neighbours (i.e. $\gamma = 0$) then only immediate neighbours are used for comparison. In other words, only pairs of individuals with $b_{ij} = 1$ are used for comparison. The normalising factor now becomes the degree of individual i, denoted by k_i, and the measure of individual experienced inequality is

$$e_i = \frac{\sum_{j \neq i, b_{ij}=1} |x_i - x_j|}{\sum_{j \neq i} \mathbb{I}_{a_{ij}=1}} = \frac{\sum_{j \neq i, b_{ij}=1} |x_i - x_j|}{k_i} \tag{2.8}$$

Let us call this i's *neighborhood average difference.*

Since, we use absolute difference as the measure of distance between nodes, the measure is symmetric. This means that both the richer and the poorer nodes on two sides of a link experience the same inequality through the link. This assumption can be relaxed and the inequality experienced because of a link can be weighed differently depending on whether the node is at the richer or poorer end of the link. In Appendix A we create such a measure and use it to create an asymmetric analogue of neighbourhood average difference. We then also aggregate it to create an asymmetric index of experienced inequality. For the rest of the paper, though, we shall follow the symmetric version shown in Equation 2.8 above.

2.2. Axioms - aggregation. The next few axioms deal with aggregating individual experiences of inequality up to a measure of societal experience of inequality.

For simplicity, let elements $a_{i,j} = 1$ or $a_{i,j} = 0$ denotes the presence or absence of a link between individuals i and j. e_i would then be equal to the neighborhood average difference of node i. Let us denote the neighborhood average difference of node i by Δ_i, such that, for what follows, $e_i = \Delta_i$.

A4. Aggregation *The aggregate experienced inequality ($\mathbb{E}$) in a society is a function of individual experienced inequality of each member of the society (e_i), and is increasing in its arguments.*

Since the individual level experienced inequality e_i is a function of the wealth distribution $\mathbf{x}$ and the social network $\mathbf{A}$, a transfer of wealth that changes $\mathbf{x}$, or the addition or deletion of links that change $\mathbf{A}$, can both change the aggregate experienced inequality ($\mathbb{E}$). Using neighborhood average difference, two corollaries that follow from the axiom put conditions on the direction of these changes.

NAD.C1 *Transferring wealth from node i to another node j reduces (increases) aggregate experienced inequality $\mathbb{E}$ if the neighbourhood average difference for the two nodes (Δ_i and Δ_j), and all nodes connected to them, all decrease (increase) as a result of the transfer.*

The formal proof for this corollary is in Appendix F. Consider Figure 2 for example. Here, when we move from panel (A) to panel (B), the underlying wealth distribution is actually becoming more equal. The Gini goes from 0.484 to 0.480. However, experienced inequality rises since the neighborhood average difference of all nodes that are affected by the transfer rises. In other words, each of these individuals is now relatively more different compared to a majority of their neighbors. This property of experienced inequality is in violation of the Pigou-Dalton transfer principle.

NAD.C2 *Adding a new link between two unconnected nodes i and j decreases (increases) aggregate experienced inequality $\mathbb{E}$ if the neighbourhood average difference for the two nodes (Δ_i and Δ_j) both decrease (increase) as a result of the addition of the link.*

The formal proof for this corollary is in Appendix F. In Figure 3, for example, moving from panel (A) to panel (B) would reduce experienced inequality because, for the two newly connected nodes, neighborhood average difference falls from $\frac{20}{2} = 10$ to $\frac{20}{3} = 6.67$.

Using neighbourhood average difference, we get two further corollaries that seem intuitive, and hence, are arguments for using this measure.

NAD.C3 *Adding a link between nodes with equal wealth can never increase experienced inequality.*

The formal proof for this corollary is in Appendix F. Adding an edge between any two nodes of equal wealth will not change the aggregate experienced differences for the individual nodes, but will increase the number of links, so that the average difference per link reduces. See Figure 3, for instance.

NAD.C4: Maximal and Minimal Inequality *Given total wealth and number of nodes, a star network with the central node having all the wealth, has the highest possible level of experienced inequality. Given total wealth and number of nodes, any network with all nodes having the same amount of wealth has the lowest possible level of experienced inequality.*

The formal proof for this corollary is in Appendix F. See Figure 4 for an example of a network where $\mathbb{E}$ is maximized for a given total wealth.

2.3. Other axioms. The fifth axiom deals with two desirable features in an index. One is that the specific location of an individual in the wealth vector $\mathbf{x}$ and the adjacency matrix $\mathbf{A}$ should not matter i.e. if one reverses the ordering in $\mathbf{x}$ and $\mathbf{A}$, the resulting aggregate experienced inequality should remain unchanged. The second feature is that the effect of an individual i's experience of inequality e_i, on the aggregate experience of inequality $\mathbb{E}$ should only depend on e_i and not on the experienced inequality of other individuals e_{-i}.

A5: Anonymity and separability *The relationship between the aggregate and individual level inequality is the same for every member of the society i.e.* $\frac{\partial \mathbb{E}}{\partial e_i}$ *is the same for all* i*, and is independent of* e_{-i}*.*

This axiom allows us to add the individual experienced inequality of all individuals to measure aggregate experienced inequality.

To our existing axioms we now add two commonly used axioms - scale and population invariance.

A6. Scale invariance: If the experienced inequality for a society with wealth distribution $\mathbf{x}$ and social network $\mathbf{A}$ is $\mathbb{E}$, then the experienced inequality for a society with the same social network $\mathbf{A}$, but wealth distribution $t\mathbf{x}$, $t > 0$, is also $\mathbb{E}$.

A7. Population invariance: Let a society with n individuals have individual experienced inequalities given by a vector $(e_1, e_2, .., e_n)$ and an aggregate experienced inequality of $\mathbb{E}$. Then a society with $2n$ individuals that have individual experienced inequalities given by a vector $(e_1, e_2, .., e_{2n})$ where $e_{n+j} = e_j$ for all $j \in 1, 2, .., n$, will also have aggregate experienced inequality $\mathbb{E}$.

An example of this would be starting with wealth distribution $\mathbf{x}$ and social network $\mathbf{A}$, and adding an identical society with the same wealth distribution and social network, but with the two networks remaining unconnected. The condition given in **A7** is more general and includes the possibility of the network simply expanding in a way such that the experience of inequality of the original individuals remain unchanged while that of the new ones are identical to that of the originals.

2.4. Index. The following index, a simple average of the individual neighbourhood average differences, normalised by twice the mean wealth, satisfies all the seven axioms.

$$\mathbb{E} = \frac{(\sum_{i=1}^{n} \Delta_i)/n}{2(\sum_{i=1}^{n} x_i)/n} \tag{2.9}$$

Interestingly, the numerator is equal to the weighted sum of link differences where the weight is the sum of inverse of the degrees of the two nodes of the link.

$$\mathbb{E} = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} (\frac{1}{k_i} + \frac{1}{k_j}) a_{i,j} |x_i - x_j|}{2 \sum_{i=1}^{n} x_i} \tag{2.10}$$

where $k_i = \sum_{j \neq i} a_{ij}$ or the degree of node i.

$\mathbb{E}$ is bounded by [0,n/2].

As we move from a sparse to a fully connected network, where all pairwise comparisons are made, there is a convergence between $\mathbb{E}$ and the experience based Gini (G) proposed in Bowles and Carlin (2020). Specifically, $\mathbb{E} = G$, for a complete network[2].

$\mathbb{E}$ is also different from the index of relative deprivation proposed most recently by Stark, Bielawski, and Falniowski (2023). Here, the measure of individual experienced inequality, e_i, includes absolute differences between i and all its neighbors, while relative deprivation only includes positive differences. Moreover, $\mathbb{E}$ is normalized such that, for a complete network, $G = \mathbb{E}$ while the aggregate index for relative deprivation is equal to $(n-1)G$.

2.5. Network multiplier of the Gini. From our discussion above, it is clear that $\mathbb{E}$ is likely to be affected by both the wealth distribution and the network structure. When $\mathbb{E}$ rises or falls, therefore, it could be caused by overall changes in the wealth distribution or changes in the network structure. One way of delineating the role of the network structure is by observing the ratio of $\mathbb{E}$ to the Gini coefficient, G.

$$r = \frac{\mathbb{E}}{G} \tag{2.11}$$

[2]The Gini coefficient is far and away the most widely used measure of inequality. It has been extensively explored and given an axiomatization (see for example, Thon (1982) and Aaberge (2001) among others).

We can call r the network multiplier of the Gini coefficient. Put another way, when we multiply r by the Gini coefficient, we get the aggregate inequality experienced on a network.

3. Experienced inequality in real-world networks

3.1. Data. We use a publicly available dataset that was initially collected by Banerjee et al. (2013) for a study on the diffusion of micro-finance loans. It contains detailed information about individuals and households in the 75 villages in Karntaka, India that participated in the study.

In each village, all households were surveyed for household wealth related information including type of roof material, number of rooms, number of beds, presence and provider of electricity and a toilet, caste, religion, presence of a political leader in the house, etc. Nearly half of the individuals were then given a more detailed survey that included questions on social connections.

The name generator questions[3] used were as follows:

(1) In your free time, whose house do you visit? (Network: visitgo)
(2) Who visits your house in his/her free time? (Network: visitcome)
(3) List relatives. (Network: rel)
(4) List non-relatives. (Network: nonrel)
(5) If you had a medical emergency and were alone at home, whom would you ask for help in getting to a hospital? (Network: medic)
(6) If you needed to borrow kerosene or rice, to whom would you go? (Network: keroricego)
(7) Who would come to you if he/she needed to borrow kerosene or rice? (Network: kerorricecome)
(8) Do you visit temple/mosque/church? Do you go with anyone else? If so, who? (Network: templecome)
(9) If you suddenly needed to borrow Rs.50 for a day, who would you ask? (Network: borrowmoney)
(10) Who do you trust enough, that if he/she needed to borrow Rs.50 for a day you would lend it to him/her? (Network: lendmoney)
(11) If you had to make a difficult personal decision, whom would you ask for advice? (Network: helpdecision)
(12) Who comes to you for advice? (Network: giveadvice)

A connection in the *all* network between two households exists if two individuals from each household are connected in any way. The 12 name generator questions listed above, along with the *all* network, give us 13 networks for each of our 75 villages. We, therefore, have data

[3]Name generator questions are meant to ask the respondent about a specific type of link, while aiming to generate a full list of names of people that the respondent comes in contact with. Collectively, they are meant to ensure that the respondent does not miss someone key that they interact with. For more details see Shakya, Christakis, and Fowler (2017) and Genkin et al. (2022).

on 975 separate networks. In this section, we compute experienced inequality ($\mathbb{E}$) and the Gini coefficient (G) on the *all* network for each village. But, where applicable, and specifically for our analysis of network properties in section 4, we compute $\mathbb{E}$, G, and other network characteristics for all 975 networks i.e. for each network in each village.

Figure 5 shows plots of three networks in one village - two of the name generator networks mentioned above and the *all* network. For our analysis of experienced inequality we exclude the singletons.

We have two potential options for our measure of household wealth. One option would be to aggregate household characteristics using principal components analysis (PCA) to create a wealth index. However, such an index would not be a linear, cardinal measure of wealth with an unambiguous lower bound value of 0.

The other option, the one we choose here, is to use a metric that is indicative of household wealth and well-being while also being linear and cardinal. Here we use the number of rooms in the house[4]. Using the number of rooms as our metric comes with two additional advantages. First, it is quite a visible feature about another person's house. So, it works well in this context where a link is a source of information about the other's wealth. Second, it correlates well with the wealth index generated using principal components analysis[5].

Using the number of rooms as an indicator of household wealth is not without drawbacks. First, we are not accounting for whether the house is rented or owned. And second, we are not accounting for the fact that some rooms may be larger than others. But, given the data we have at our disposal, this seems like the most reasonable way to proceed. Nonetheless, the results that we present here are robust to using the PCA wealth index. Our main results are replicated using the PCA wealth index in Appendix B. With the number of rooms as an indicator of wealth we get a mean wealth of 2.361, a mean neighborhood average difference of 1.252 (see Table 1), and 0.278 as our overall Gini coefficient.

For the measurement of experienced inequality, we operationalize the connectedness between individuals, and therefore each individual's comparator set through regular interactions. This is only one way to instantiate the comparator set. Individuals may compare themselves to others based on the news, on publicly available data, in social media interactions, etc. Furthermore, the valence of comparisons may differ depending on the network. Comparisons within relative

[4]The minimum number of rooms reported for a household is 0. This is not immediately intuitive but is understandable given the survey. The survey question underlying this variable was: "How many rooms does your house have?" Different people might interpret "rooms" differently. Some might answer with the number of bedrooms, others with the number of separate rooms or partitions, etc. So, some people that have just one room for the entire house might say 0. Of the 14904 households in the dataset, 201 or about 1.35% of the households report having 0 rooms. Of this, 177 also report having no latrine.

[5]See Figure 6 for a distribution of number of rooms in a house in our dataset and Figure 7 for its correlation with a wealth index generated through PCA.

and friend networks may be very different than one based on, say business networks. While we recognize the shortcomings of the networks we use here, our purpose in the empirical section, is mainly to identify how an inequality measure that relaxes the assumptions of full information and full connectedness can be generated on any given network. We are also interested in seeing the difference such a measure makes relative to the standard measure of inequality. A more accurate term for this exercise may be 'interaction based inequality' for 'experienced inequality'.

We must also acknowledge that just because an individual interacts with another, it does not imply that they have accurate information about the other's wealth. People often form beliefs about the wealth of others. And they may rely on a lot of factors to form these beliefs including but not limited to visible cues. Rinn et al. (2023), for example, find that participants in their study thought that they could identify rich people based on the following dimensions: "luxury consumption, expensive hobbies, spontaneous spending,... and specific possessions." One might be more likely to observe these cues through social and economic interactions. But, social and economic interactions are not the only way for people to observe such cues. Moreover, the way that these cues are interpreted may be subject to bias for two reasons. First, people may act with an intention to signal a certain level of wealth (Veblen and Mills 1899). And, second, people may hold inaccurate stereotypes about the ways in which visible cues translate to actual wealth (see Kappes, Gladstone, and Hershfield (2021), for example). Here, we assume that people's beliefs about the wealth of those in their network are true.

Moreover, we use households as opposed to individuals as nodes. This is because of the nature of the dataset we use. It consists of a census of households. But, only individuals from 50% of the households were given a detailed survey that included questions on social network connections. Because of this survey methodology, we do not have data on all of the individuals from the households that were not surveyed. The measures might differ if we were to look at the individual level, however. This is for two reasons. First, different individuals in the same household may have different levels of wealth. A younger individual may have less accumulated wealth compared to an older individual, for example. Second, the network itself might be different. People in the same household may have different social connections. If more data were available our measure can readily be applied to situations where the nodes are individuals as opposed to households. Our measure is agnostic to the type of node.

3.2. Comparing experienced inequality and the Gini coefficient. As seen in the top panel of Table 2, in the 75 villages in our dataset, $\mathbb{E}$ ranges from 0.184 to 0.443, with a mean of 0.265. The Gini coefficient in the 75 villages ranges from 0.187 to 0.427, with a mean of 0.269. r ranges from 0.770 to 1.353, with a mean of 0.983. $r > 1$ in 25 villages, $r < 1$ in 50 villages and in no village is $r = 1$. Figure 8 shows kernel density estimates for r, $\mathbb{E}$, and G.

Figure 9 shows the scatter plot of the Gini coefficient and $\mathbb{E}$ for the *all* network. We can see that $\mathbb{E}$ and G share a strong, positive, linear relationship. It can also be seen that the fitted line is almost as steep as the 45-degree line suggesting that on average a higher Gini coefficient is associated with increased experienced inequality.

In Figure 10, we rank the 75 villages in our dataset in increasing order of the Gini coefficient from bottom to top on the left, and in increasing order of $\mathbb{E}$ on the right. A line connects the same village on the left and the right. The higher the r, the lighter the shade of the line. Even though, Figure 9 shows a strong, positive, linear relationship between $\mathbb{E}$ and G, it becomes clear from this figure that a village that is relatively more unequal might be *experiencing* relatively less inequality depending on the network structure.

Figure 11 shows mean r in various types of networks generated by the name generator questions mentioned in subsection 3.1 above. Notice how r is lowest for the templecompany and rel (relative) networks where individuals are likely to interact with other individuals most like themselves. On the other hand, r is greater for the borrowmoney, nonrel (non-relative connections), and helpdecision networks where individuals are more likely to name people higher in social status. Although these differences exist, they are not statistically significant differences. In the following section when we analyze the impact of network properties on r, therefore, we use a pooled dataset with all types of networks from all villages.

In the lower panel of Table 2 we can see that experienced inequality in all 975 networks in our dataset ranged from 0.119 to 0.514 with a mean of 0.257. The network multiplier of the Gini ranged from 0.566 to 1.401 with a mean of 0.961.

4. Homophily and experienced inequality

In this section, we explore how homophily in networks could affect $\mathbb{E}$. To do this, we first conceptually explain how two types of homophily - wealth homophily and caste homophily - could be related to experienced inequality for a given wealth distribution (subsection 4.1). We then see if the relationships we predict hold in real world networks (subsection 4.2 and subsection 4.3).

4.1. Definitions and relationships.

Wealth homophily. It is possible that people actively choose to associate with others who have similar levels of wealth i.e. the network displays wealth homophily. The relationship between experienced inequality and wealth homophily can be intuitively anticipated. The higher the level of wealth homophily, the more likely that links with lower wealth difference will be formed thus lowering both individual and aggregate experienced inequality. A formal proof is presented here using an augmented Erdos-Renyi network formation process, showing that for a given network density, an increase in wealth homophily will reduce experienced inequality.

We define wealth homophily through the random link formation process described by the equation below.

$$P(a_{i,j} = 1) = \delta - h\frac{|x_i - x_j| - \Delta_c}{\Delta_c} \tag{4.1}$$

Here, δ is the density, h is the wealth homophily parameter and Δ_c is the average wealth difference across all possible pairs.

To estimate this we run a linear regression with $a_{i,j}$ as the dependent variable and the normalised wealth difference as the independent variable. The coefficient estimate is

$$\begin{aligned}
\hat{h} &= -\frac{\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}(a_{i,j}-\delta)\left(\frac{|x_i-x_j|-\Delta_c}{\Delta_c}\right)}{\sigma_c^2} \\
&= -\frac{\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}a_{i,j}(|x_i-x_j|-\Delta_c)-\delta\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}(|x_i-x_j|-\Delta_c)}{\Delta_c\sigma_c^2} \\
&= -\frac{\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}a_{i,j}|x_i-x_j|-\frac{\Delta_c n\bar{k}}{2}}{\Delta_c\sigma_c^2}
\end{aligned} \tag{4.2}$$

Here, σ_c^2 is the variance of the normalised wealth difference and $\bar{k}$ is mean degree. Hence, for networks where the degree distribution does not diverge too much from the mean, $k_i \approx \bar{k}$, we have

$$\mathbb{E} = \frac{\frac{2}{k}\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}a_{i,j}|x_i-x_j|}{2\sum_{i=1}^{n}x_i} = \frac{\frac{2}{k}\Delta_c\left[\frac{n\bar{k}}{2}-\hat{h}\sigma_c^2\right]}{2\sum_{i=1}^{n}x_i} \tag{4.3}$$

Hence, experienced inequality decreases as wealth homophily $\hat{h}$ increases. However, in reality, the degree distribution can vary substantially from the average degree. To check the robustness of our analytical result, we run simulations of randomly generated networks and show that the relationship still holds (see Appendix C).

Caste homophily. It is well documented[6] that people are more likely to form links within social groups like race, ethnicity, or caste. We can intuitively anticipate that the relationship between group homophily and experienced inequality will depend on within-group inequality. If within-group inequality is lower than the overall inequality then increasing links within the group, at the expense of those between groups, may decrease experienced inequality. On the other hand, if within-group inequality is higher than the overall inequality, increasing links within the group, at the expense of those between groups, may increase experienced inequality.

For the purposes of this section, we define group homophily (caste homophily in this context), as the proportion of within-group links as a share of the total number of links[7].

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[6]See Davidson (2018), for example.

[7]This is a very simple measure of within-group homophily, which is also influenced by relative group size. We use it for its simplicity that helps in deriving our analytical result. For more sophisticated measures that are not affected by group size, see Karimi and Oliveira (2023) and Sajjadi, Martin-Gutierrez, and Karimi (2024).

Proposition: *In a society with two distinct groups, an increase in group homophily, while keeping the degree distribution unchanged, will lead to an increase in experienced inequality if the average within group Gini, weighted by the wealth share of the groups, is larger than the overall Gini.*

The proof is in Appendix D. In the rest of the section, when we refer to "average within-group inequality", we are referring to the average within group Gini coefficient, weighted by the wealth share of the groups. In other words,

$$\text{Average within-group inequality} = G_1 w_1 + G_2 w_2 \tag{4.4}$$

where G_i and w_i are the within-group Gini coefficient and wealth share of group i.

4.2. Data. In order to examine the relationships between wealth homophily, caste homophily, and r, we use a pooled dataset of all 13 networks in each of the 75 villages. For each network in each village we compute experienced inequality, the Gini coefficient, and the network multiplier. For each network in each village we also estimate wealth homophily and caste homophily. Hence, we get 975 observations of all these parameters from these networks, shown in Table 2.

We estimate wealth homophily in each network in each village using Equation 4.2. From Table 2 we can see that wealth homophily ranges from -0.023 to 0.034 with a mean of 0. Figure 12(A) shows the relationship between wealth homophily and r. As anticipated, greater wealth homophily is associated with a smaller r.

In order to estimate caste homophily, we divide the village into two groups. The first group consists of households in the the largest caste in a village. The second group consists of all other households. Caste homophily is then measured as the number of same-group links as a proportion of all links in any network. From Table 2, it can be seen that the share of households belonging to the largest caste varies from 0.246 to 1 with a mean of 0.622. Caste homophily varies from 0.542 to 1 with a mean of 0.831.

From the proposition in section 4.1, we can see that the relationship between $\mathbb{E}$ and caste homophily is likely to vary based on how within-caste inequality compares to the overall Gini coefficient. If average within-caste inequality is greater than the overall Gini coefficient, increasing caste homophily will increase $\mathbb{E}$. If average within-caste inequality is less than the overall Gini coefficient, increasing caste homophily will decrease $\mathbb{E}$. In order to check if this condition is met, we also compute the within-caste Gini and wealth share for each caste-group. From Table 2, it can be seen that the wealth share of the largest caste varies from 0.235 to 1 with a mean of 0.639.

Figure 12(B) shows the relationship between caste homophily and r when average within-caste inequality is less than the overall Gini coefficient. Figure 12(C) shows the relationship between caste homophily and r when average within-caste inequality is greater than the overall Gini coefficient. As anticipated, greater caste homophily is associated with a smaller r in the former case i.e. when average within-caste inequality is less than the overall Gini coefficient. However in the latter case, i.e. when average within-caste inequality is greater than the overall Gini coefficient, caste homophily and r do not share a strong positive or negative relationship.

In Appendix E, we address the unanticipated result in Figure 12(C). We do this in two parts. First, we show that the relationship between r and caste homophily is stronger when the average within-caste inequality is farther away from the Gini coefficient. And, one possible explanation for the flat relationship in Figure 12(C) is that in most networks where average within-caste inequality is greater than the overall Gini, average within-caste inequality is still very close to the Gini. Second, we come up with a corrected measure of caste homophily that accounts for the effect of varying group size and show that the direction of the relationships are still as anticipated, in a linear regression.

4.3. Regression results. In Table 3 we estimate how the Gini coefficient, wealth homophily, and caste homophily are associated with experienced inequality (in models (2) to (4)). We also examine how they impact r (in model (5)).

Models (1), (2), (3) and (4) suggest that increasing Gini is associated with a less than proportional increase in $\mathbb{E}$. A unit increase in the Gini coefficient is associated with a 0.972 increase in $\mathbb{E}$ in Model (1) and 0.963, 0.963, and 0.931 in models (2), (3), and (4), respectively. Model (2) controls for wealth homophily and includes network fixed effects; Model (3) controls for caste homophily; and Model (4) controls for wealth homophily and caste homophily, and includes network fixed effects.

Models (2) and (4) show that even after holding the Gini coefficient and caste homophily constant, increasing wealth homophily ($\hat{h}$) is associated with a decline in experienced inequality. A unit increase in wealth homophily is associated with a 4.9 unit decline in $\mathbb{E}$ in Model (2) and 4.479 unit decline in Model (4). Additionally, increasing wealth homophily is associated with a 17.849 unit decline in r according to Model (5) where we control for caste homophily and include network fixed effects.

Based on our proposition in section 4.1, increased caste homophily should be associated with greater experienced inequality in villages where average within-caste inequality is greater than the overall Gini coefficient. For the purposes of this regression, we create a binary variable ($\mathbf{I}$) that takes a value of 1 for networks where the average within-caste inequality is greater than the overall Gini coefficient. Based on our hypothesis, the coefficient on caste homophily should be negative and the coefficient on the interaction term between caste homophily and $\mathbf{I}$ should be positive. Since caste homophily might increase just as function of one group getting larger relative to the other, we also control for population shares.

From Model (4), we can see that increasing caste homophily is associated with a decrease in experienced inequality by 0.022 points when within inequality is less than the overall Gini coefficient. On the other hand, caste homophily increases experienced inequality by 0.028 points when within inequality is greater than the overall Gini coefficient. The same relationship holds for r. Caste homophily decreases the network multiplier by 0.108 points when within inequality is less than the overall Gini coefficient and increases it by 0.09 points when the opposite is true.

5. Conclusion and discussion

In this paper, we have explored the ways in which structures of association between individuals affect their experience of inequality. Since individuals meet and engage with more people like themselves, the absence of ties between individuals with disparate wealth can lower the experience of inequality relative to standard measures. Interestingly, we find that this may not always be the case - there are instances in which the experienced inequality in social networks is larger than what might be measured using standard approaches. Caste homophily might increase experienced inequality when within-caste inequality is greater than the Gini coefficient and decrease experienced inequality otherwise.

The proposed measure of experienced inequality captures how the connections between people of different wealth and incomes affect the experience of inequality at local and aggregate levels. In doing that, it makes it possible to ask and answer several policy relevant questions. Five examples come to mind. First, it has been established that local, neighborhood or community level inequality informs people's perceptions of overall country-level inequality. It is this perception that informs their policy preferences (see Hauser and Norton (2017) and Knell and Stix (2020), for example). Our measure extends this to the social networks on which people interact. $\mathbb{E}$, is therefore, likely to be a strong predictor of people's perceptions of inequality and their policy preferences. Second, Chetty et al. (2022) find that "economic connectedness" ie. the connectedness between those with low versus high socio-economic status is "among the strongest predictors" of upward economic mobility "to date". Our measure can be used to characterize the level of economic connectedness in different networks. Third, Nishi et al. (2015) find that it is not just inequality but wealth visibility that predicts cooperative behavior on networks. Our measure allows us to quantify wealth visibility both at the individual and at the aggregate network level. Fourth, our paper also contributes to recent efforts to explore the role of social networks in shaping inequality patterns. Espín-Noboa et al. (2022), for instance, show how, and to what extent, well-known ranking and recommendation algorithms produce inequality when applied to social networks. Finally, several recent studies have brought a network lens to economic and social inequality (see for example, Avin et al. (2015) and McMillan (2022)). In these contexts, our work can be used to measure inequality in a way that is cognizant of the overall distribution as well as the network topology.

We conclude here by reflecting on some implications of our line of reasoning that can be taken up in subsequent work. First, we have not tried to characterize our measure. A useful theoretical extension may be to expand the range of axiomatic underpinnings to narrow down the properties and forms of experienced inequality. These axioms will clearly have to extend beyond those central to inequality measurement.

Second, more work may be done to understand the interpretation of links between people and its relationship to inequality as experienced. There is by now, a long standing psychological interpretation of inequality as aggregated envy. It is unclear to us that such an interpretation is necessary or appropriate in all cases. Depending on context, linked nodes that possess different income or wealth may interpret these as information about opportunities as much as anything else. A poorer individual who has been segregated with only other poor individuals may view a link with a person outside his or her group not as envy but as information that can support their decision making.

Third and finally, the most important and interesting extension will be to take this to other real world data. An ideal dataset would contain information on individuals' perception of their neighbors' wealth and their beliefs about overall levels of inequality. This would allow us to analyze what network connections and status goods are relatively more relevant for social comparisons. There are also some network connections that are missing in the dataset we use. For example, we do not have information on which individuals work together and which individuals live geographically close to each other. Moreover, richer community level information would have been very useful. For instance, survey questions about levels of trust and life satisfaction would have allowed us to say more about the implications of higher or lower experienced inequality. In this dataset we only have a single cross section. Further applications to richer and time variant real world data may provide greater insight. For example, do societies which display increasing experienced inequality over time also exhibit patterns of increasing conflict, or decreasing propensities to undertake collective action? Does this differ from what might be predicted using more standard measures of inequality? Going forward, our measure may be used to address such questions of critical importance.

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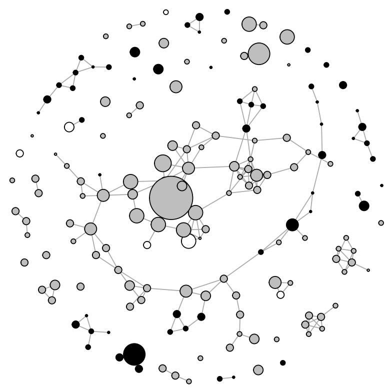
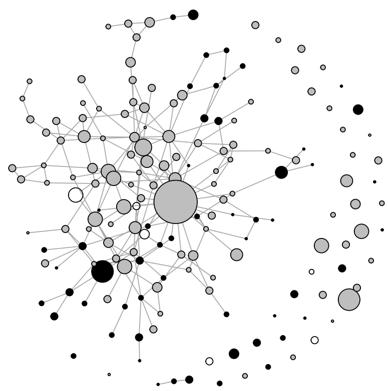
Figures



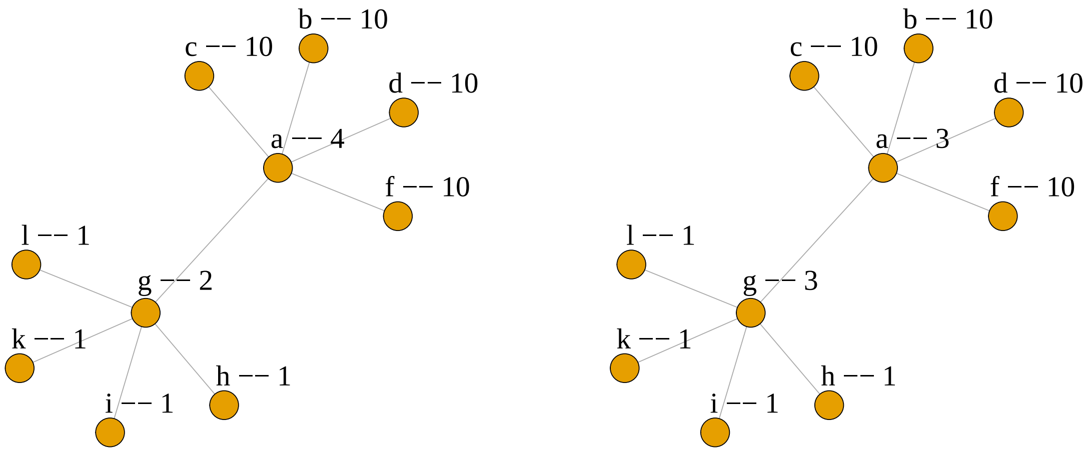
(A) Low experienced inequality ($\mathbb{E} = 0.255$) for a given wealth distribution ($G = 0.277$)



(B) High experienced inequality ($\mathbb{E} = 0.388$) for a given wealth distribution ($G = 0.277$)

Figure 1. Two networks in the same village

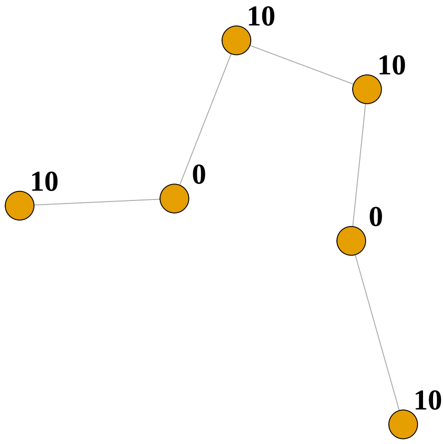
Note: The size of the node is positively correlated to household wealth. The color of the node is associated with the caste of the household. Gray nodes indicate households from the largest caste in the village. Black nodes indicate households from all other castes. The figure in the top panel depicts the relative/kinship network and the figure in the lower panel depicts the help-decision network for the same village. For the same underlying wealth distribution, experienced inequality is larger in the network in Panel (B) than in the network in Panel (A).



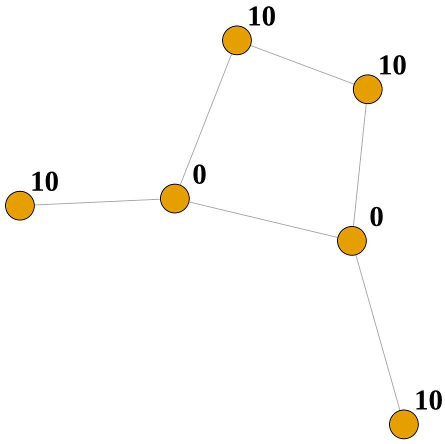
(A) (B)

Figure 2. A transfer of $1 from the node **a** to node **g** decreases the Gini coefficient but increases experienced inequality.

Note: The label next to each node indicates its name for reference and its wealth. A transfer of $1 from node **a** to node **g** decreases the Gini coefficient but increases experienced inequality because **a**'s, **g**'s, and all their neighbors' neighborhood average differences rise. **a**'s neighborhood average difference goes from 5.2 to 5.6. **g**'s neighborhood average difference goes from 1.2 to 1.6. **l**, **k**, **i**, and **h**'s neighborhood average differences go from 1 to 2. **c**, **b**, **d**, and **f**'s neighborhood average differences go from 6 to 7. This property of experienced inequality violates the Pigou-Dalton transfer principle.



(A)



(B)

FIGURE 3. Experienced inequality declines when the new link is added.

Note: The label next to each node indicates its wealth. Experienced inequality declines when the new link is added because the neighborhood average difference for each of the newly connected nodes declines from $\frac{20}{2} = 10$ in panel (A) to $\frac{20}{3} = 6.67$ in panel (B).

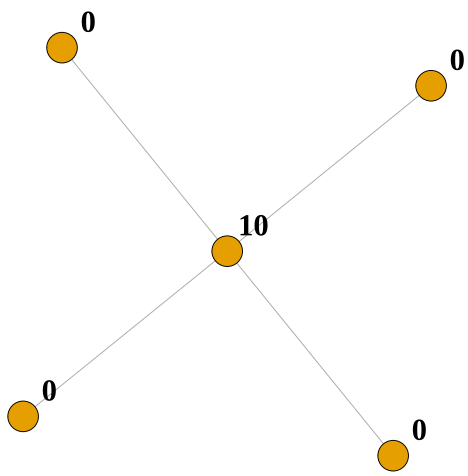
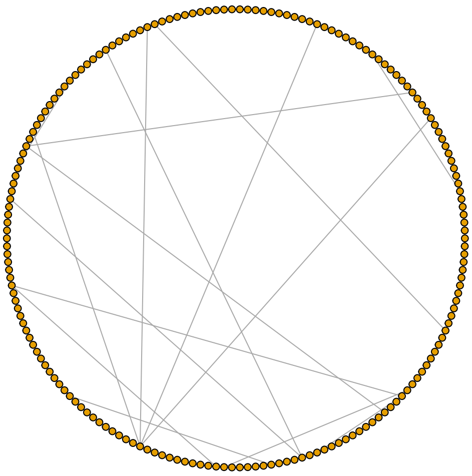
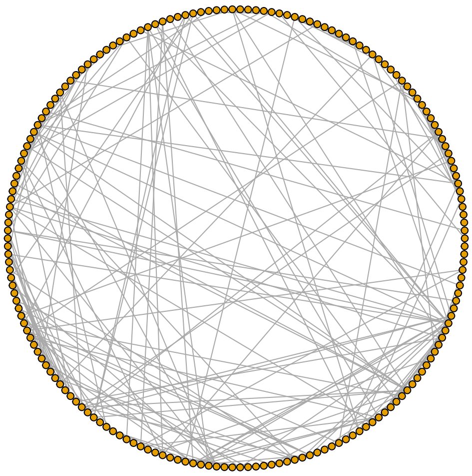


FIGURE 4. An example of a star network where experienced inequality is maximized for a given total wealth and number of nodes

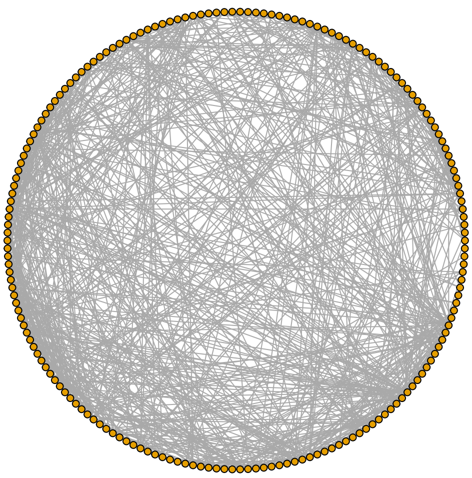
Note: The label next to each node indicates its wealth.



(A) Temple company



(B) Borrow money



(C) All

FIGURE 5. Temple company, borrow money, and the *all* network of a sample village in the dataset.

Note: This figure depicts the temple company network, the money borrowing network, and the *all* network for a sample village in the dataset. Nearly half of the individuals in each village were given a detailed survey about social connections. They were asked name generator questions such as: “Who visits your hour house in his/her free time?” “Who would come to you if he/she needed to borrow kerosene or rice?” Two households are connected if two individuals from these households are connected.

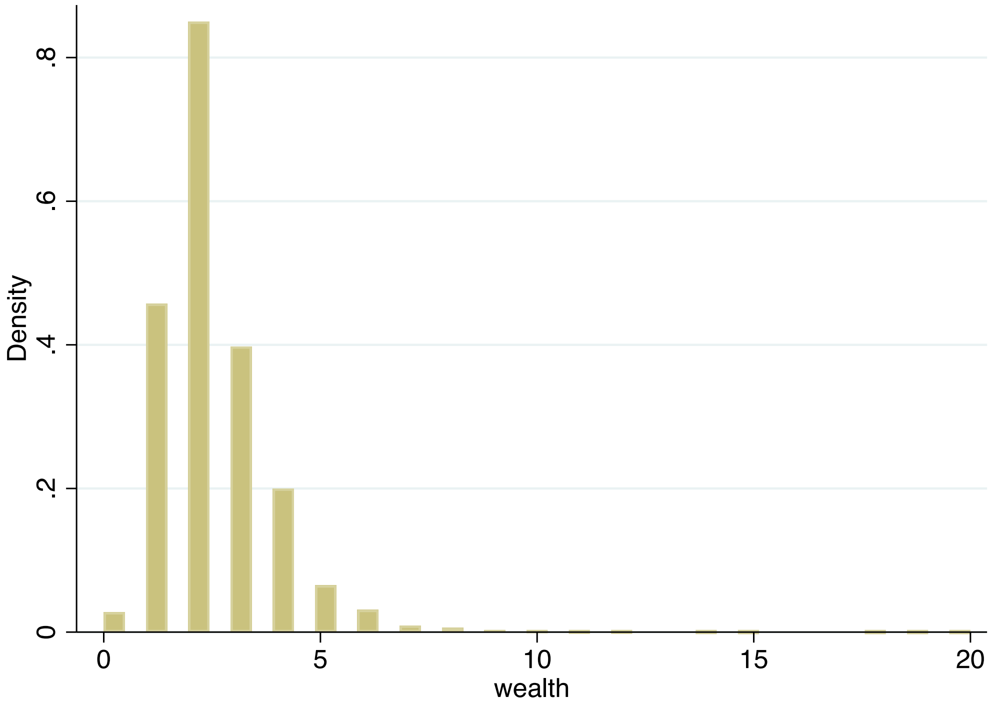


Figure 6. Histogram of number of rooms (used as an indicator of household wealth).

Note: This figure shows the histogram for the number of rooms in a house among 14904 households in the dataset. We use the number of rooms as our indicator of household wealth.

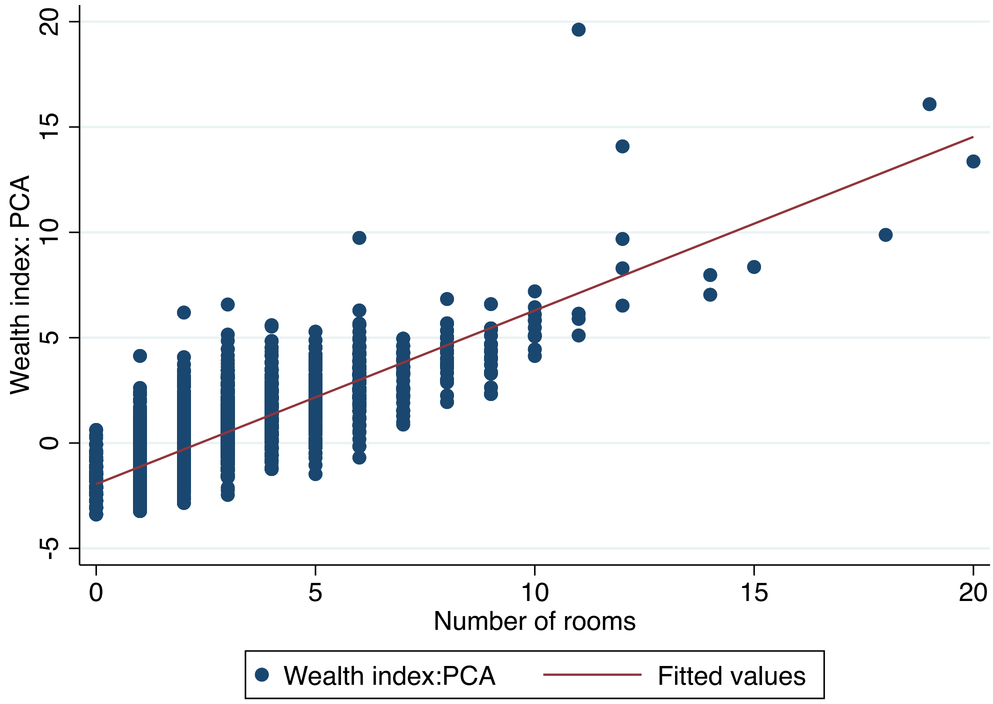
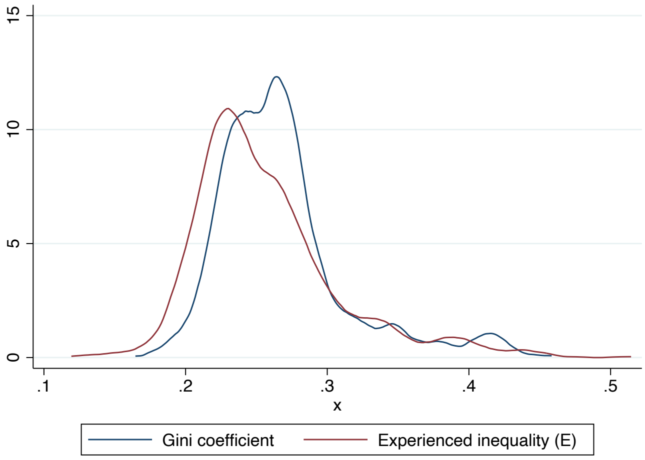
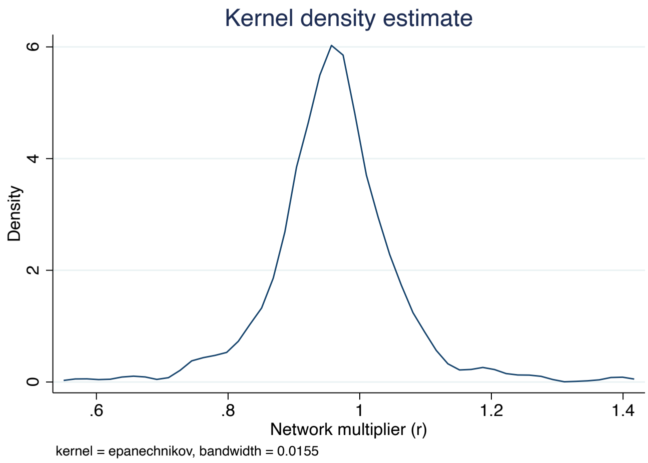


Figure 7. Scatter plot of number of rooms and a wealth index aggregated through principal components analysis (PCA)

Note: The wealth index variable was constructed using principal components analysis on six observable household characteristics: type of roof, number of rooms, number of beds, type of electricity connection, presence of a latrine, and ownership of the house. Summary statistics for these disaggregated indicators are in Table B.1.



(A)



(B)

Figure 8. Distribution of the Gini coefficient, $\mathbb{E}$, and r

Note: Smoothed density curves for the Gini coefficient, experienced inequality ($\mathbb{E}$, on the left), and the network multiplier of the Gini (r, on the right.) The density curve for the Gini coefficient includes 75 observations for 75 villages. The experienced inequality and network multiplier density curves include observations for each of the 13 networks in each of the 75 villages. They therefore depict the density curves over 975 observations.

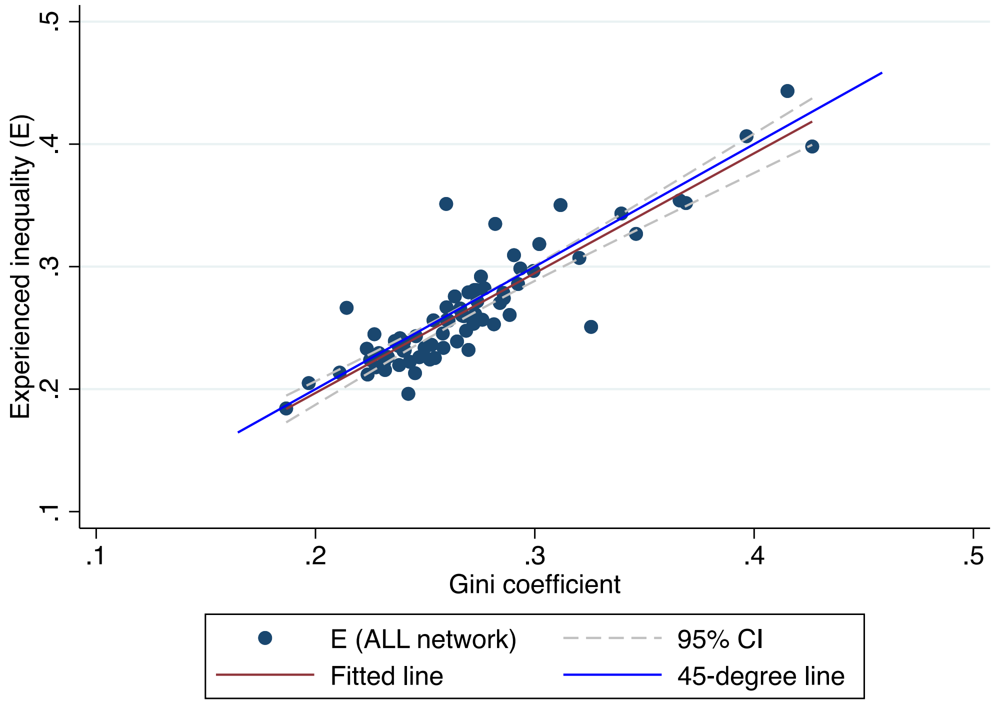


Figure 9. $\mathbb{E}$ versus the Gini coefficient.

Note: The figure depicts a scatter plot for the Gini coefficient, G, and the experienced inequality, $\mathbb{E}$ on the *all* network for 75 villages. The red line is the fitted line, the dashed lines depict the 95% confidence intervals, and the blue line is the 45-degree line.

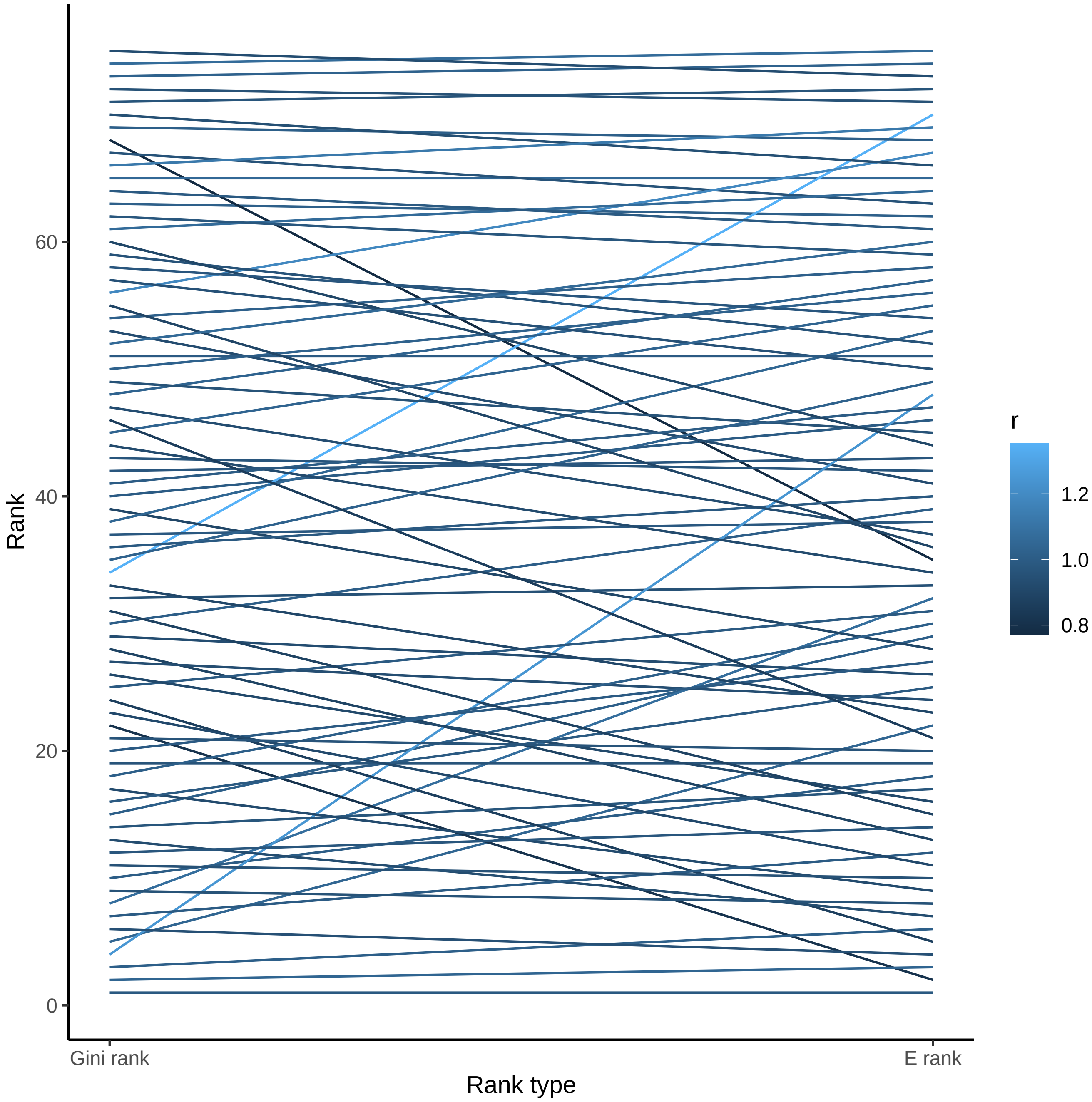


Figure 10. Rank change in villages - Gini coefficient vs experienced inequality

Note: On the left, the 75 villages in our dataset are ranked in increasing order of Gini coefficient from bottom to top. On the right, the same 75 villages are ranked in increasing order of experienced inequality on the *all* network from bottom to top. A line connects the same village. The lighter the shade of blue of the line, the higher the network multiplier of the Gini, r.

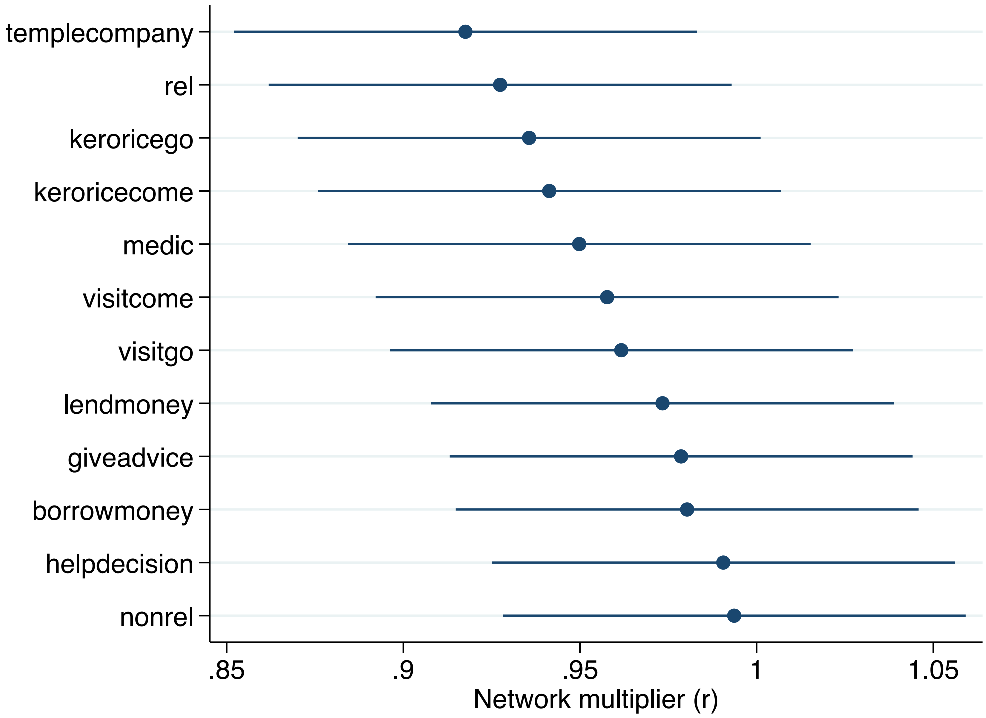
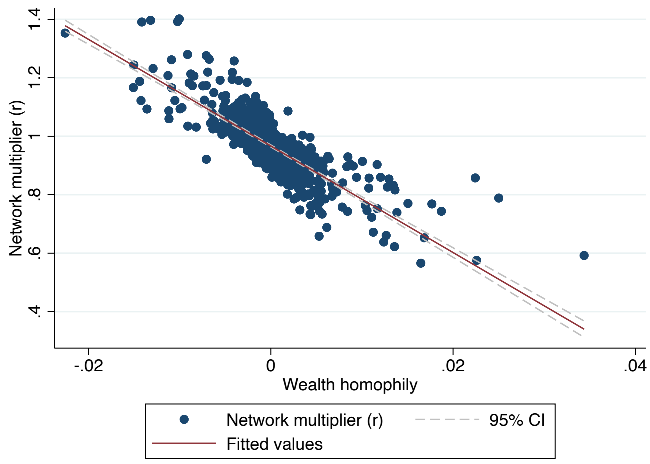
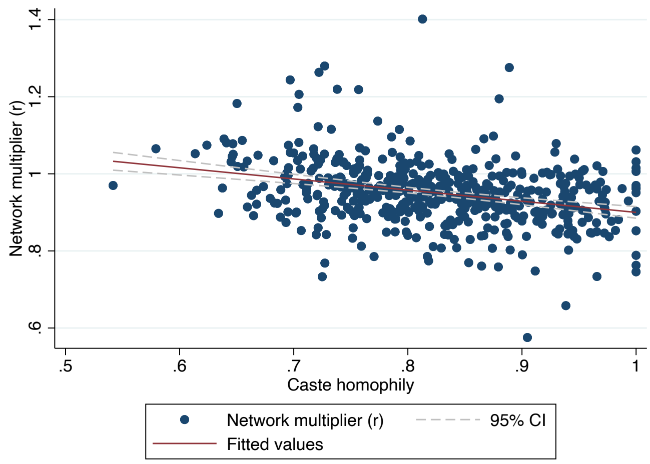


Figure 11. r by type of network.

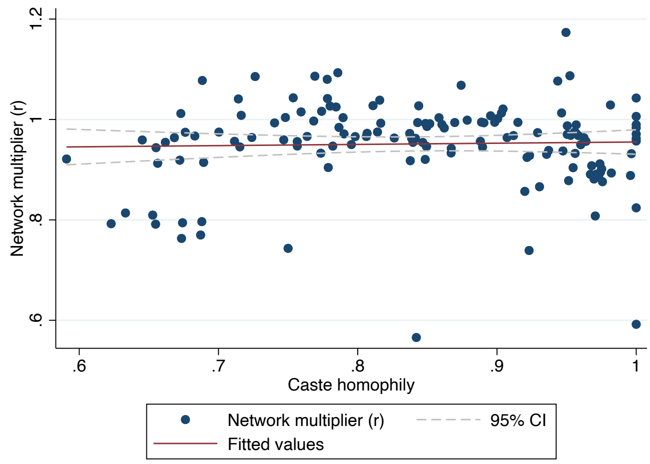
Note: This figure shows the mean network multiplier of the Gini coefficient, r, by type of network with 95% confidence intervals. The mean for "templecompany", for example, is the mean r for 75 "templecompany" networks - one in each of the 75 villages.



(A)



(B) Avg within-caste inequality < Gini



(C) Avg within-caste inequality > Gini

FIGURE 12

Note: The three panels in this figure depict the network multiplier of the Gini (r) on the vertical axis and wealth homophily ($\hat{h}$, in panel A) and caste homophily (s, in panels B and C) on the horizontal axis for real-world networks. $\hat{h}$ is computed using Equation 4.2. s is computed as the proportion of same-caste links in any network. $\hat{h}$, s, and r are all computed separately for each network in each village. Panel B includes networks in which average within-caste inequality is less than the overall Gini coefficient. Panel C includes networks in which average within-caste inequality is greater than the overall Gini coefficient. The red line is the fitted line, and the dashed lines represent the 95% confidence interval.

Tables

TABLE 1. Summary statistics - household

	Mean	SD	Min	Max	N
Household wealth	2.361	1.315	0	20	14904
Degree	8.972	7.278	0	90	14904
Neighborhood average difference	1.252	0.898	0	16.25	14165

Note: Household wealth refers to the number of rooms in each household. Degree for each household is computed on the *all* network. Neighborhood average difference is computed using the definition in Equation 2.8. We do not compute neighborhood average difference for households whose own wealth information is not available and for households with a degree of 0.

Table 2. Summary statistics - Village x Network characteristics

	Mean	SD	Min	Max	N
Village					
Gini coefficient	0.269	0.045	0.187	0.427	75.000
Experienced inequality - ALL	0.265	0.050	0.184	0.443	75.000
Network multiplier of Gini - ALL	0.983	0.084	0.770	1.353	75.000
Village size	188.867	57.149	75.000	341.000	75.000
Share of HHs in largest caste	0.621	0.182	0.305	1.000	50.000
Wealth share of largest caste	0.638	0.194	0.308	1.000	50.000
Village x Network					
Experienced inequality	0.257	0.052	0.119	0.514	975.000
Network multiplier of Gini (r)	0.961	0.094	0.566	1.401	975.000
Number of links	343.362	240.563	7.000	2015.000	975.000
Wealth homophily	0.000	0.004	-0.023	0.034	975.000
Caste homophily	0.831	0.095	0.542	1.000	650.000
Share of HHs in largest caste	0.622	0.184	0.246	1.000	650.000
Wealth share of largest caste	0.639	0.196	0.235	1.000	650.000

Note: This table shows the summary statistics for each village (top panel) and for each network in each village (lower panel). The Gini coefficient, the village size, the share of households belonging to the largest caste, and the wealth share of the largest caste are computed for each village after excluding singleton nodes. Experienced inequality ($\mathbb{E}$) and the network multiplier of the Gini (r) in the top panel are computed for the *all* network in each village. Separately, in the bottom panel, experienced inequality, the network multiplier of the Gini, number of links, wealth homophily ($\hat{h}$), caste homophily (s), share of households belonging to the largest caste, and wealth share of the largest caste are computed for each network in each village after excluding singleton nodes. $\hat{h}$ is computed as in Equation 4.2 and s is computed as the share of same-caste links in a network.

TABLE 3. Regression analysis

	$\mathbb{E}$				r
	(1)	(2)	(3)	(4)	(5)
Constant	-0.003	0.001	0.057	0.025	1.040
	(0.013)	(0.008)	(0.028)	(0.012)	(0.035)
Gini coefficient	0.972	0.963	0.963	0.931	
	(0.049)	(0.032)	(0.078)	(0.043)	
Wealth homophily		-4.900		-4.479	-17.849
		(0.354)		(0.413)	(1.951)
Caste homophily			-0.073	-0.022	-0.108
			(0.020)	(0.010)	(0.041)
I			-0.065	-0.040	-0.161
			(0.040)	(0.016)	(0.069)
I*Caste homophily			0.079	0.050	0.198
			(0.045)	(0.019)	(0.080)
Observations	975	975	650	650	650
Network fixed effects	N	Y	N	Y	Y
Population share control	-	-	N	Y	Y

Note: This table shows the estimates of the regression as mentioned in subsection 4.3. The dataset for real-world networks includes all networks in all villages. The first four models are estimated with experienced inequality, $\mathbb{E}$, as the dependent variable. The fifth model is estimated using the network multiplier of the Gini coefficient, r, as the dependent variable. Wealth homophily ($\hat{h}$) is computed as per Equation 4.2 and caste homophily, s, is computed as the share of same-caste links in a network. The indicator variable **I**, is a binary variable that takes the value of 1 when within-caste inequality is greater than the overall Gini coefficient. **I***Caste Homophily is the interaction term. We do not compute caste homophily for the 25 villages (and therefore the 375 networks – 13 for each village) where caste information was not available. Standard errors (in parentheses) are clustered at the level of the village.

Appendix

Appendix A. An alternative measure for individual experienced inequality

Here we discuss an alternative measure of individual experienced inequality and the consequences of using it for the aggregate measure.

In the neighbourhood average difference measure, we take the absolute value of the difference in wealth between an individual and their immediate neighbour. We may want to allow for a different weight being allotted to the difference depending on whether the neighbour in question is richer or poorer than the individual.

Following from the inequality averse utility function proposed in Fehr and Schmidt (1999), we can write the asymmetric neighbourhood average difference as

$$\Delta_i^a = \frac{\sum_{j \neq i} ((1-\alpha)max(x_i - x_j, 0) + (1+\alpha)max(x_j - x_i, 0))a_{i,j}}{\sum_{j \neq i} a_{i,j}}$$

For $\alpha = 0$, this collapses to the symmetric version we have used. For $0 < \alpha < 1$, this incorporates the asymmetry where individuals care much more about those differences where they are poorer than where they are richer. And finally for $\alpha = 1$, this becomes a measure of individual experienced deprivation where we only count the differences where the individual is poorer than the neighbour in question.

The aggregate measure in this case would

$$\mathbb{E}^a = \frac{(\sum_{i=1}^{n} \Delta_i^a)/n}{2(\sum_{i=1}^{n} x_i)/n} = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} ((\frac{1}{d_i} + \frac{1}{d_j}) + \alpha(1 - 2\mathbf{I}_{x_i < x_j}(x_i, x_j))(\frac{1}{d_i} - \frac{1}{d_j}))a_{i,j}|x_i - x_j|}{2\sum_{i=1}^{n} x_i} \tag{A.1}$$

Here, $\mathbf{I}_{x_i < x_j}(x_i, x_j)$ is an indicator function that takes the value 1 if $x_i < x_j$, and 0 otherwise. The index also converges to the Gini for a complete network as the degree for all nodes is the same.

Appendix B. Composite wealth index

Table B.1. Summary statistics - wealth indicators

		Mean	SD	Min	Max	N
Rooftype	Thatch	0.023	0.151	0	1	14904
	Tile	0.324	0.468	0	1	14904
	Stone	0.304	0.460	0	1	14904
	Sheet	0.193	0.394	0	1	14904
	RCC	0.117	0.321	0	1	14904
Number of rooms		2.361	1.315	0	20	14904
Number of beds		0.849	1.331	0	50	14904
Electricity	Private	0.617	0.486	0	1	14904
	Government	0.306	0.461	0	1	14904
	None	0.077	0.266	0	1	14904
Latrine	Owned	0.261	0.439	0	1	14904
	Common	0.006	0.075	0	1	14904
	None	0.733	0.442	0	1	14904
House owned		0.897	0.304	0	1	14904

Note: This table summarizes the six main indicators that were used to construct the composite wealth index. For rooftype, electricity, latrine, and house owned, the mean indicates the proportion in each category. Private, Government, and None indicate proportion of houses that have privately provided, government provided, and no electricity connection.

TABLE B.2. Summary statistics - Village x Network characteristics

	Mean	SD	Min	Max	N
Village					
Gini coefficient	0.225	0.029	0.144	0.330	75.000
Experienced inequality - ALL	0.205	0.025	0.138	0.266	75.000
Network multiplier of Gini - ALL	0.912	0.067	0.763	1.112	75.000
Village size	188.867	57.149	75.000	341.000	75.000
Share of HHs in largest caste	0.621	0.182	0.305	1.000	50.000
Wealth share of largest caste	0.641	0.196	0.308	1.000	50.000
Village x Network					
Experienced inequality	0.196	0.027	0.113	0.304	975.000
Network multiplier of Gini (r)	0.885	0.087	0.391	1.264	975.000
Number of links	343.362	240.563	7.000	2015.000	975.000
Wealth homophily	0.004	0.006	-0.034	0.068	975.000
Caste homophily	0.831	0.095	0.542	1.000	650.000
Share of HHs in largest caste	0.622	0.184	0.246	1.000	650.000
Wealth share of largest caste	0.642	0.199	0.217	1.000	650.000

Note: This table replicates the summary statistics in Table 2 but for household wealth measured by a composite wealth index aggregated through principal components analysis (PCA). Summary statistics for each village are in the top panel and for each network in each village are in the lower panel. The Gini coefficient, the village size, the share of households belonging to the largest caste, and the wealth share of the largest caste are computed for each village after excluding singleton nodes. Experienced inequality ($\mathbb{E}$) and the network multiplier of the Gini (r) in the top panel are computed for the *all* network in each village. Separately, in the bottom panel, experienced inequality, the network multiplier of the Gini, number of links, wealth homophily ($\hat{h}$), caste homophily (s), share of households belonging to the largest caste, and the wealth share of the largest caste are computed for each network in each village after excluding singleton nodes. $\hat{h}$ is computed as in Equation 4.2 and s is computed as the share of same-caste links in a network.

TABLE B.3. Regression analysis

	$\mathbb{E}$				r
	(1)	(2)	(3)	(4)	(5)
Constant	0.049	0.051	0.058	0.034	0.934
	(0.012)	(0.009)	(0.027)	(0.016)	(0.048)
Gini coefficient	0.661	0.726	0.705	0.797	
	(0.055)	(0.039)	(0.060)	(0.039)	
Wealth homophily		-2.270		-2.071	-9.633
		(0.199)		(0.201)	(1.017)
Caste homophily			-0.022	-0.014	-0.057
			(0.021)	(0.014)	(0.065)
I			-0.070	0.001	0.070
			(0.029)	(0.024)	(0.099)
I*Caste homophily			0.085	0.005	-0.048
			(0.036)	(0.027)	(0.111)
Observations	975	975	650	650	650
Network fixed effects	N	Y	N	Y	Y
Population share control	-	-	N	Y	Y

Note: This table replicates the estimates of the regression as in Table 3 but for wealth measured by a composite wealth index aggregated through principal components analysis (PCA). The dataset for real-world networks includes all networks in all villages. The first four models are estimated with experienced inequality, $\mathbb{E}$, as the dependent variable. The fifth model is estimated using the network multiplier of the Gini coefficient, r, as the dependent variable. Wealth homophily ($\hat{h}$) is computed as per Equation 4.2 and caste homophily, s, is computed as the share of same-caste links in a network. The indicator variable **I**, is a binary variable that takes the value of 1 when within group inequality is greater than the overall Gini coefficient. **I***Caste Homophily is the interaction term. We do not compute caste homophily for the 25 villages (and therefore the 375 networks – 13 for each village) where caste information was not available. Standard errors (in parentheses) are clustered at the level of the village.

Appendix C. Simulated networks

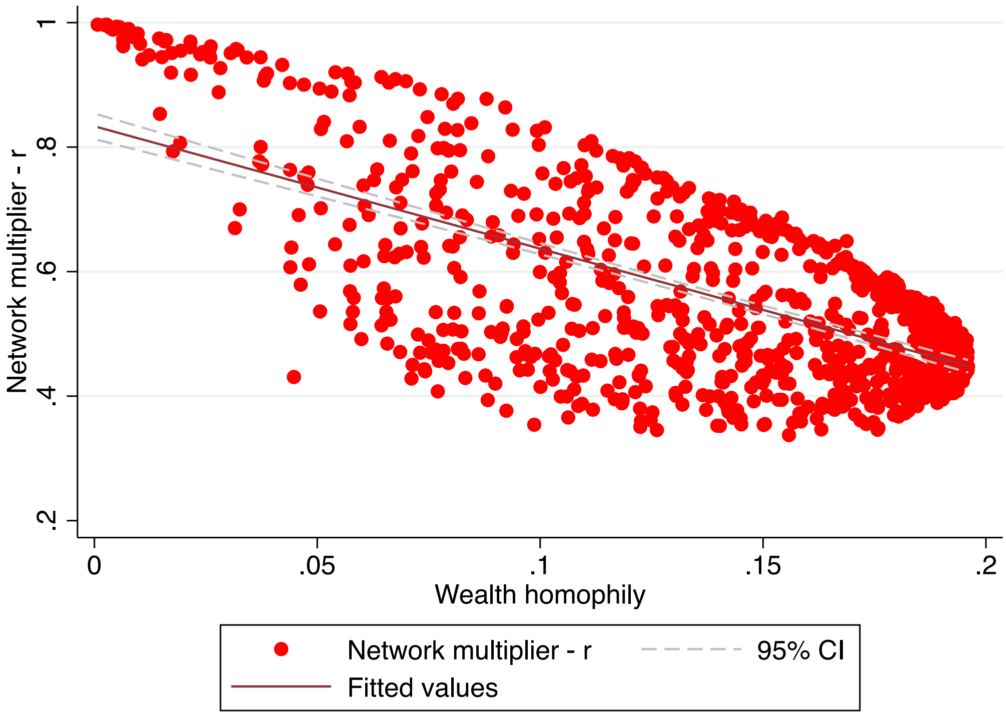


FIGURE C.1. Wealth homophily and the network multiplier of the Gini coefficient, r, on simulated networks

Note: This figure depicts the network multiplier of the Gini (r) on the vertical axis and the wealth homophily ($\hat{h}$) on the horizontal axis for 975 simulated networks. The networks are simulated using an Erdos-Renyi-type algorithm with the probability of link formation given by Equation 4.1. We use randomly drawn values for h between 0 and 1. We assume there are 300 nodes. The wealth of the individual nodes are drawn from a Pareto distribution with shape parameter $a = 2$ and mean income $m = 0.2$. $\hat{h}$ and r are all computed for each network separately. The red line is the fitted line, and the dashed lines represent the 95% confidence interval.

Appendix D. Proof of proposition - caste homophily

We consider a network where each node belongs to one of two groups S_1 and S_2. Let us consider the numerator of Equation 2.10 only. It can be re-written as follows:

$$\sum_{i\in S_1}\sum_{j\in S_1}(\frac{1}{k_i}+\frac{1}{k_j})a_{i,j}|x_i-x_j|+\sum_{i\in S_2}\sum_{j\in S_2}(\frac{1}{k_i}+\frac{1}{k_j})a_{i,j}|x_i-x_j|+\sum_{i\in S_1}\sum_{j\in S_2}(\frac{1}{k_i}+\frac{1}{k_j})a_{i,j}|x_i-x_j|$$

Consider two between-group links that connect nodes a_1 and b_1 in group S_1 to nodes a_2 and b_2 in group S_2, such that a_1-b_1, and a_2-b_2 are not connected. Now assume that the links a_1-a_2 and b_1-b_2 get deleted and instead a_1-b_1 and a_2-b_2 are formed. This would increase caste-homophily, while preserving the degrees of all nodes.

The numerator of $\mathbb{E}$ will increase by $(\frac{1}{k_{a_1}}+\frac{1}{k_{b_1}})|x_{a_1}-x_{b_1}|+(\frac{1}{k_{a_2}}+\frac{1}{k_{b_2}})|x_{a_2}-x_{b_2}|$ and decrease by $(\frac{1}{k_{a_1}}+\frac{1}{k_{a_2}})|x_{a_1}-x_{a_2}|+(\frac{1}{k_{b_1}}+\frac{1}{k_{b_2}})|x_{b_1}-x_{b_2}|$.

If we assume that

(i) the degree distribution of the two groups is the same;

(ii) there is no wealth homophily; and

(iii) the degree and wealth distributions are independent,

we can abstract away from the weight on each link. In that case, increased homophily will increase the expected value of $\mathbb{E}$ if

$$E[|x_{a_1}-x_{b_1}|]+E[|x_{a_2}-x_{b_2}|]>E[|x_{a_1}-x_{a_2}|]+E[|x_{b_1}-x_{b_2}|] \tag{D.1}$$

Here, $E[.]$ denotes the expected value. The left hand side of the inequality in Equation D.1 can be written as

$$\begin{aligned}
&E[|x_{a1}-x_{b1}|]+E[|x_{a2}-x_{b2}|]\\
=&\frac{\sum\limits_{i\in S_1}\sum\limits_{j\in S_1}|x_{i1}-x_{j1}|}{n_1(n_1-1)}+\frac{\sum\limits_{i\in S_2}\sum\limits_{j\in S_2}|x_{i2}-x_{j2}|}{n_2(n_2-1)}\\
=&G_1 2\bar{x}_1+G_2 2\bar{x}_2
\end{aligned}$$

The right hand side of the inequality in Equation D.1 can be written as

$$\begin{aligned}
&E[|x_{a_1}-x_{a_2}|]+E[|x_{b_1}-x_{b_2}|]\\
=&2\frac{\sum\limits_{i\in S_1}\sum\limits_{j\in S_2}|x_{i1}-x_{j2}|}{n_1 n_2}
\end{aligned}$$

Equation D.1 can therefore be rewritten as

$$G_1\bar{x_1} + G_2\bar{x_2} > \frac{\sum\limits_{i\in S_1}\sum\limits_{j\in S_2}|x_{i1} - x_{j2}|}{n_1 n_2} \tag{D.2}$$

Now, recall that

$$\begin{aligned}
G &= \frac{\sum\limits_{i=1}^{n-1}\sum\limits_{j=i+1}^{n}|x_i - x_j|}{\frac{n(n-1)}{2}2\bar{x}} \\
&= \frac{\sum\limits_{i\in S_1=1}^{n_1-1}\sum\limits_{j\in S_1=i+1}^{n_1}|x_{1i} - x_{1j}|}{\frac{n(n-1)}{2}2\bar{x}} + \frac{\sum\limits_{i\in S_2=1}^{n_2-1}\sum\limits_{j\in S_2=i+1}^{n_2}|x_i - x_j|}{\frac{n(n-1)}{2}2\bar{x}} + \frac{\sum\limits_{i\in S_1=1}^{n_1}\sum\limits_{j\in S_2=1}^{n_2}|x_i - x_j|}{\frac{n(n-1)}{2}2\bar{x}} \\
&= G_1\frac{n_1(n_1-1)\bar{x_1}}{n(n-1)\bar{x}} + G_2\frac{n_2(n_2-1)\bar{x_2}}{n(n-1)\bar{x}} + \frac{\sum\limits_{i\in S_1}\sum\limits_{j\in S_2}|x_{i1} - x_{j2}|}{n_1 n_2}\frac{n_1 n_2}{n(n-1)\bar{x}}
\end{aligned} \tag{D.3}$$

For large n_1, n_2, and n, $n_1 - 1 \approx n_1$, $n_2 - 1 \approx n_2$ and $n - 1 \approx n$. Let the population shares be given by $\frac{n_1}{n} = p_1$, $\frac{n_2}{n} = p_2$. Let the wealth shares be $\frac{\sum_{i\in S_1}^{n_1} x_{1i}}{\sum_i^n x_i} = w_1$ and $\frac{\sum_{i\in S_2}^{n_2} x_{2i}}{\sum_i^n x_i} = w_2$. This would imply that $\frac{\bar{x_1}}{\bar{x}} = \frac{w_1}{p_1}$ and $\frac{\bar{x_2}}{\bar{x}} = \frac{w_2}{p_2}$. Equation D.3 can be rewritten as:

$$\begin{aligned}
G &= G_1p_1w_1 + G_2p_2w_2 + \frac{\sum\limits_{i\in S_1}\sum\limits_{j\in S_2}|x_{i1} - x_{j2}|}{n_1 n_2}\frac{p_1p_2}{\bar{x}} \\
\textit{Or,}\quad (G - G_1p_1w_1 - G_2p_2w_2)\frac{\bar{x}}{p_1p_2} &= \frac{\sum\limits_{i\in S_1}\sum\limits_{j\in S_2}|x_{i1} - x_{j2}|}{n_1 n_2}
\end{aligned} \tag{D.4}$$

From Equation D.2 and Equation D.4, we know that experienced inequality will increase with group homophily if

$$\begin{aligned}
&G_1\bar{x_1} + G_2\bar{x_2} > (G - G_1p_1w_1 - G_2p_2w_2)\frac{\bar{x}}{p_1p_2} \\
\textit{Or,}\quad &G_1\frac{\bar{x_1}}{\bar{x}}p_1p_2 + G_2\frac{\bar{x_2}}{\bar{x}}p_1p_2 > G - G_1p_1w_1 - G_2p_2w_2 \\
\textit{Or,}\quad &G_1w_1p_2 + G_2w_2p_1 > G - G_1p_1w_1 - G_2p_2w_2 \\
\textit{Or,}\quad &G_1w_1 + G_2w_2 > G
\end{aligned}$$

Appendix E. Caste homophily and experienced inequality

E.1. Explaining Figure 12(C). In Figure E.1 and Figure E.2 we can see that in most networks where average within-caste inequality is greater than the overall Gini coefficient, average within-caste inequality is still very close to the Gini coefficient. Specifically, Figure E.1, shows a scatterplot of the Gini coefficient on the horizontal axis and the average within-caste inequality on the vertical axis. The red line is the 45-degree line where the Gini coefficient is equal to the average within-caste inequality. We can see from this figure that points above the red line are far fewer and much closer to the red line than points below. Figure E.2 shows a histogram of the ratio of within-caste inequality to the Gini coefficient. Again, notice that the distribution is relatively left skewed with values below 1 being much farther away from 1 than values above 1.

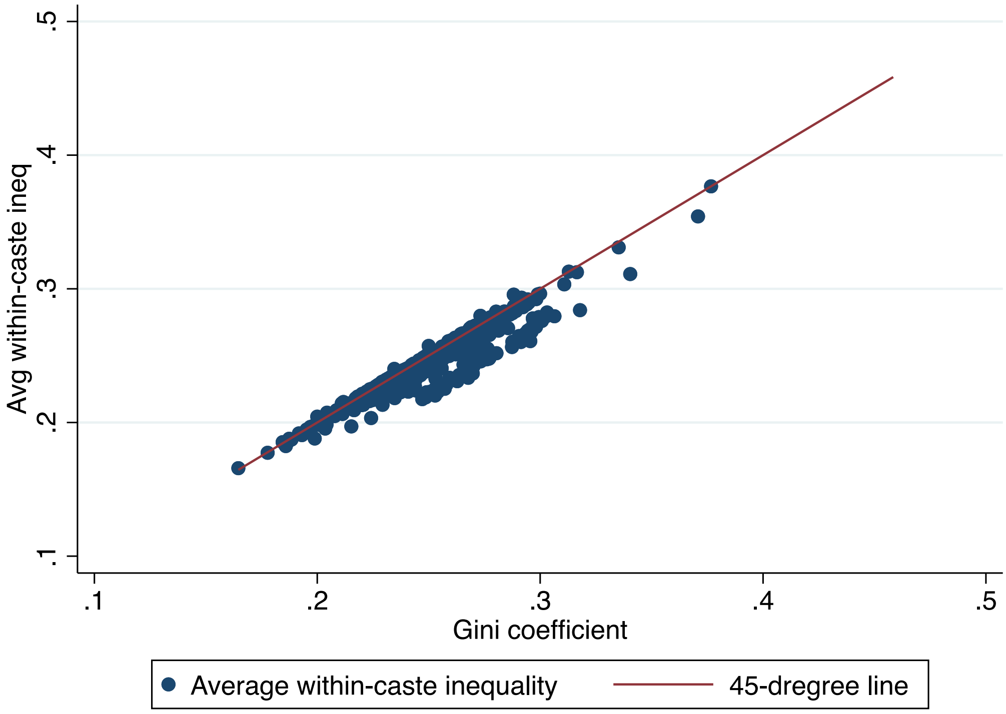


FIGURE E.1. Gini coefficient vs average within-caste inequality

Note: This figure depicts the Gini coefficient on the horizontal axis and the average within-caste inequality on the vertical axis for 650 networks for which caste data is available. The average within-caste inequality is calculated as in Equation 4.4.

Figure E.3 depicts the predicted marginal effects of caste homophily on r at various levels (0.85, 0.9, 0.95, 1, 1.05) of the ratio between average within-caste inequality and the Gini coefficient. The marginal effects are calculated by regressing the network multiplier of the Gini coefficient, r, on (a) caste homophily, (b) the ratio of within-caste inequality to the Gini coefficient, and (c) an interaction term of caste homophily and the ratio of average within-caste inequality to the Gini coeffient. We also control for wealth homophily, population share, and

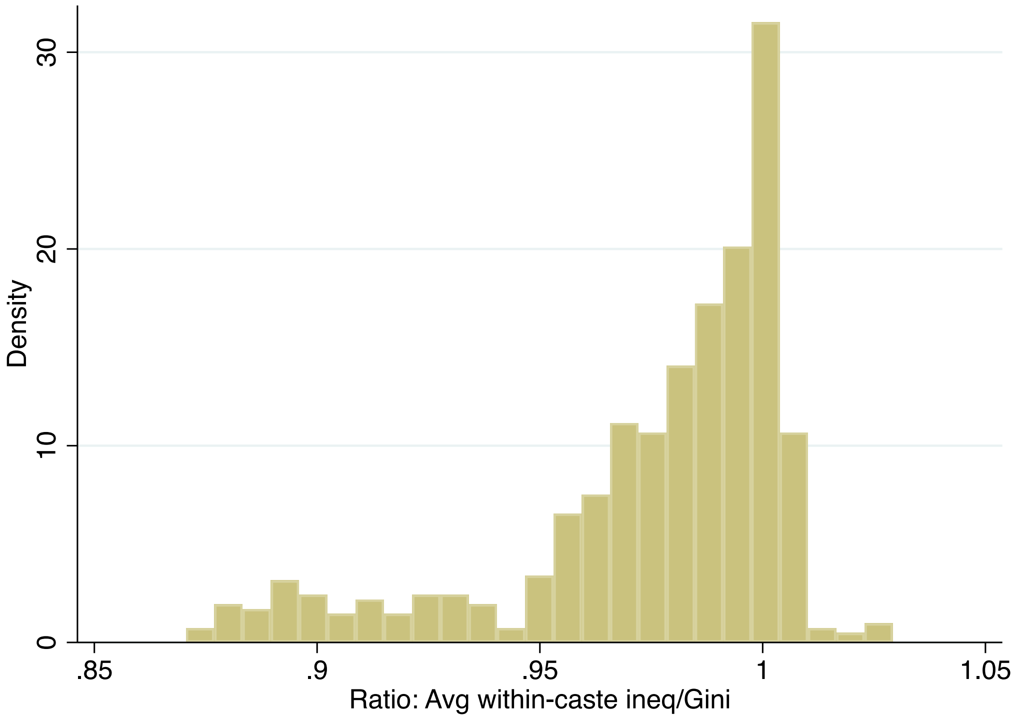


FIGURE E.2. Histogram of the ratio of average within-caste inequality to the Gini coefficient

Note: This figure depicts a histogram of the ratio of the average within-caste inequality to the Gini coefficient. The average within-caste inequality is calculated as in Equation 4.4. Values greater than 1 are networks where average within-caste inequality is greater than the Gini coefficient and values less than 1 are networks where the average within-caste inequality is less than the Gini coefficient.

network fixed effects. We can see the relationship between caste homophily and the network multiplier of the Gini coefficient r is stronger, when average within-caste inequality is farther away from the Gini coefficient.

Therefore, one potential explanation for the flat relationship between caste homophily and r in Figure 12(C) is that in networks where average within-caste inequality is greater than the Gini coefficient, average within-caste inequality is still very close to the Gini coefficient. And, in such cases, the relationship between caste homophily and r tends to be weak.

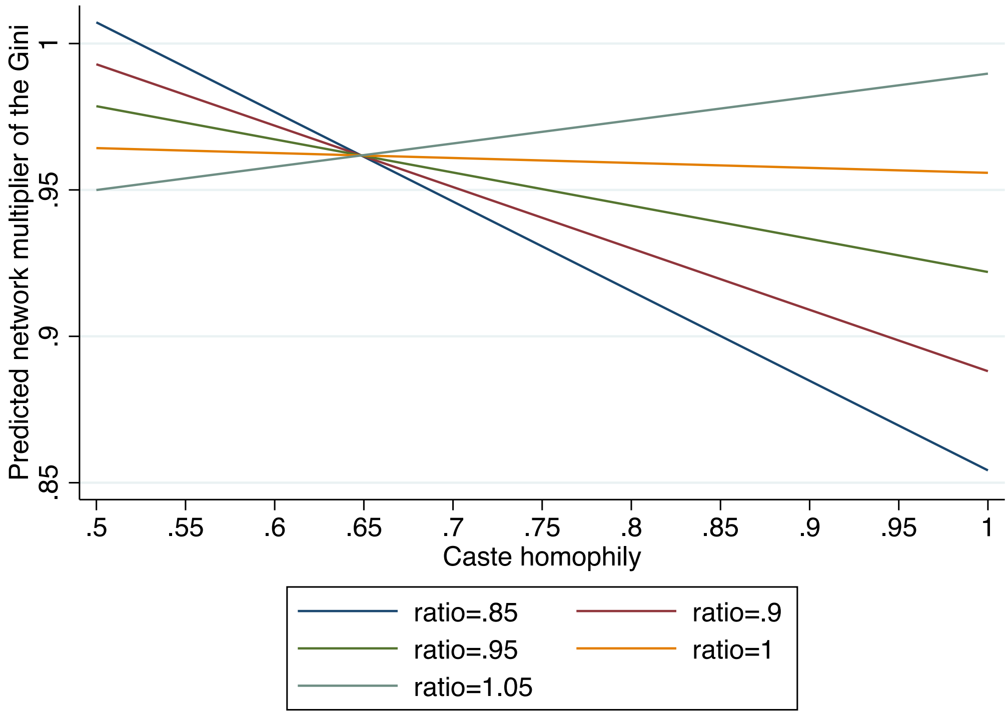


FIGURE E.3. Predicted marginal effects of caste homophily on r at various levels of the ratio of average within-caste inequality to the Gini coefficient

Note: This figure depicts the predicted marginal effects of caste homophily on r at various levels (0.85, 0.9, 0.95, 1, 1.05) of the ratio between average within-caste inequality and the Gini coefficient. The marginal effects are calculated by regressing the network multiplier of the Gini coefficient, r, on (a) caste homophily, (b) the ratio of within-caste inequality to the Gini coefficient, and (c) an interaction term of caste homophily and the ratio of average within-caste inequality to the Gini coeffient. We also control for wealth homophily, population share, and network fixed effects as in Table 3.

E.2. Corrected caste homophily measure. In this section, we check the robustness of our results to a corrected measure of caste homophily that accounts for the effect of varying group size. We correct our measure by dividing our original measure by what the measure would have been if links had been randomly formed. In Equation E.1 below, the numerator is our original measure. In the denominator, n_1 is the number of households (nodes) in one group and n_2 is the number of households in the other group. n is the total number of nodes in the network.

$$s' = \frac{\left[\dfrac{\text{Number of same caste links in network}}{\text{Number of links in network}}\right]}{\left[\dfrac{\dfrac{n_1(n_1-1)}{2} + \dfrac{n_2(n_2-1)}{2}}{\dfrac{n(n-1)}{2}}\right]} \tag{E.1}$$

Table E.1 replicates the estimates of the regression as in Table 3, Column (4) and (5) but with a corrected caste homophily measure. It can be seen that, our main results are robust to the corrected caste homophily measure. The signs of the coefficients on caste homophily and the interaction term are as predicted although statistically significant only in some cases.

Table E.1. Regression analysis

	r		$\mathbb{E}$	
	(1)	(2)	(3)	(4)
Constant	0.997	0.970	0.014	0.009
	(0.025)	(0.026)	(0.011)	(0.010)
Gini coefficient			0.934	0.927
			(0.045)	(0.045)
Wealth homophily	-18.024	-18.105	-4.504	-4.532
	(1.936)	(2.089)	(0.421)	(0.445)
Caste homophily	-0.027	-0.010	-0.006	-0.001
	(0.017)	(0.017)	(0.004)	(0.004)
I	-0.033	-0.013	-0.008	-0.003
	(0.039)	(0.038)	(0.009)	(0.009)
I*Caste homophily	0.028	0.013	0.007	0.003
	(0.029)	(0.027)	(0.007)	(0.007)
Observations	650	650	650	650
Network fixed effects	N	Y	N	Y

Note: This table replicates the estimates of the regression as in Table 3, Column (4) and (5) but with a corrected caste homophily measure that accounts for group size effects. The dataset for real-world networks includes all networks in all villages. The first two models are estimated using the network multiplier of the Gini coefficient, r, as the dependent variable. The the third and fourth models are estimated with experienced inequality, $\mathbb{E}$, as the dependent variable. Wealth homophily ($\hat{h}$) is computed as per Equation 4.2 and caste homophily, s', is computed as in Equation E.1 above. The indicator variable $\mathbf{I}$, is a binary variable that takes the value of 1 when average within-caste inequality is greater than the overall Gini coefficient. $\mathbf{I}^*$Caste Homophily is the interaction term. Standard errors (in parentheses) are clustered at the level of the village.

Appendix F. Proofs for NAD.C1, NAD.C2, NAD.C3, & NAD.C4

NAD.C1 *Transferring wealth from node i to another node j reduces (increases) aggregate experienced inequality $\mathbb{E}$ if the neighbourhood average difference for the two nodes (Δ_i and Δ_j), and all nodes connected to them, all decrease (increase) as a result of the transfer.*

Proof: Let nodes i and j be connected to k_i and k_j other nodes. So the total number of nodes whose neighbourhood average difference will get affected due to a transfer of wealth from i to j are either $k_i + k_j + 2$ or $k_i + k_j + 1$, depending on whether i and j are connected to each other. All other nodes in the network will remain unaffected. Hence, the corollary follows from the axiom. If all the affected nodes have a decrease in neighborhood average difference, the aggregate experienced inequality must decrease, and vice versa.

NAD.C2 *Adding a new link between two unconnected nodes i and j decreases (increases) aggregate experienced inequality $\mathbb{E}$ if the neighbourhood average difference for the two nodes (Δ_i and Δ_j) both decrease (increase) as a result of the addition of the link.*

Proof: The addition or deletion of a link will affect the neighbourhood average difference of only the two nodes it connects. The neighbourhood average difference of all other nodes remains the same. Hence if both the nodes experience a decrease (increase) in neighbourhood average difference and all other nodes remain unaffected, then it follows from the axiom that the aggregate experienced inequality must decrease (increase).

NAD.C3 *Adding a link between nodes with equal wealth can never increase experienced inequality.*

Proof: Consider two unconnected nodes i and j such that $x_i = x_j$. Their neighbourhood average differences are given by $\Delta_i = S_i/k_i$ and $\Delta_j = S_j/k_j$ respectively, where S_i and S_j are the respective sum of the link differences that i and j have with their neighbours, and k_i and k_j are their respective degrees.

Now if they are connected, the numerator of the expressions for neighbourhood average differences increases by $|x_i - x_j| = 0$, while the denominator increases by 1. $\Delta_i' = S_i/(k_i+1) \leq \Delta_i$ and $\Delta_j' = S_j/(k_j + 1) \leq \Delta_j$ (the equality is for the case where S_i or S_j is 0). Since, both neighbourhood average differences can either decrease or remain the same, following **NAD.C2**, aggregate experienced inequality cannot increase. The deletion case can be proved similarly.

NAD.C4: Maximal and Minimal Inequality *Given total wealth and number of nodes, a star network with the central node having all the wealth, has the highest possible level of experienced* *inequality. Given total wealth and number of nodes, any network with all nodes having the same amount of wealth has the lowest possible level of experienced inequality.*

Proof: The first part follows from corollaries **NAD.C1** and **NAD.C3**. For a star network - with all the wealth concentrated with the central node - transferring wealth or adding an edge can never increase experienced inequality further. If we transfer wealth from the central node to any other node, the neighbourhood average difference of every node in the network will fall and hence aggregate experienced inequality will fall. Any possible addition of edge will always be between nodes with equal (zero) wealth. Hence, following **NAD.C3**, experienced inequality can never increase (it will actually decrease).

The second part is obvious from the expression for aggregate experienced inequality $\mathbb{E}$, where if every node has equal wealth, $\mathbb{E} = 0$, which is the minimum value that the expression can take.